On the Formalities of Discriminability and Probability

with Application to Dynamical Systems

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Preface

This thesis presents the Observable Dynamics Program: a derivation of probabilistic and predictive structure from the primitive notion of discriminability.

The standard approach to probability begins with a probability space $(\Omega, \mathcal{F}, P)$ and studies what can be observed from it. This work inverts that order. We begin with what an observer can distinguish — a structured family of queries — and ask what that structure forces. The answer, developed across six papers, is that compatible observable distinctions force the emergence of probability, predictive kernels, operator dynamics, and computational approximation.

The six papers are presented here as chapters. Each is a self-contained article. Between them are interstitial chapters that carry the philosophical and mathematical thread connecting one paper to the next. A reader interested only in the technical results may read the article chapters directly. A reader interested in the foundational arc should read the interstitial chapters first.

The thesis has the following structure. Chapter 1 establishes the foundation: what an observer is, what they cannot avoid knowing, and the structure that their coherent commitments force. Chapters 2–6 develop the consequences: probability, predictive state, operator dynamics, and computational instantiation. The interstitial chapters explain why each transition is necessary rather than merely convenient.

Chapter 1

Discriminability and the Origin of the σ-Algebra

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Abstract

Stone's representation theorem identifies a Boolean algebra $\mathcal{E}$ with the clopen algebra of a compact totally-disconnected space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ with the Baire σ-algebra of $S(\mathcal{E})$. Under this correspondence, the classical question — when does a charge on $\mathcal{E}$ extend to a σ-additive measure on $\sigma(\mathcal{E})$? — becomes: when does a finitely-additive measure on the clopens of $S(\mathcal{E})$ extend to a Borel measure? The answer for a single algebra is well known: continuity at $\emptyset$, equivalently tightness on the Stone space.

This paper asks the question for a *directed system* of Boolean algebras: a preordered family $\{(\mathcal{E}_i, \ell_i)\}_{i\in\mathcal{I}}$, each connected by surjective refinement maps with compatible charges. The Stone spaces $\{S(\mathcal{E}_i)\}$ form a projective system, and the extension question becomes whether the charges propagate coherently to σ-additive measures throughout.

We show three things. First, the gap $\sigma(\mathcal{E}) \setminus \mathcal{E}$ is not a defect: a level with no gap admits no nontrivial refinement above it, and $\sigma(\mathcal{E}) = \mathcal{E}$ implies algebraic triviality of the refinement order. Second, index-layer coherence conditions alone — sequential upper-directedness and compatibility — do not force σ-additivity; this confirms, in the system setting, that no finitary structural condition suffices, consistent with the classical single-algebra result. The counterexample is a compatible family on a sequentially upper-directed system in which every level permanently assigns unit mass to a sequence of events the system as a whole sees as empty. Third, we identify the correct characterisation: a family of finitely-additive charges extends to σ-additive measures at every level if and only if it is *collectively exhaustive* — a valuation-layer condition requiring that every vanishing sequence is witnessed somewhere in the refinement order. Collective exhaustion is the algebraic substitute for the tightness condition on the Stone space, requiring no topology on the outcome spaces.

1.1 Introduction

A Boolean algebra $\mathcal{E}$ of events, together with a finitely-additive charge $\ell : \mathcal{E} \to [0, 1]$, is the primitive data of a discriminability structure. Under Stone's representation theorem [5], $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected space $S(\mathcal{E})$, and $\sigma(\mathcal{E})$ is the Baire σ-algebra of $S(\mathcal{E})$. The extension question — when does ℓ extend to a σ-additive measure on $\sigma(\mathcal{E})$? — becomes: when is the charge on the clopens tight enough to extend to the full Baire σ-algebra? The extension criterion (Theorem 1.8) gives the answer: continuity at $\emptyset$, in the sense of Daniell–Stone [1, 6], is necessary and sufficient. No finitary condition on $\mathcal{E}$ alone can supply this: the Stone space of a non-compact Boolean algebra carries charges that do not extend, regardless of any algebraic constraint [2]. The question is whether something beyond the single algebra can.

This paper studies the extension question for a *directed system* of Boolean algebras: a preordered family $\{(\mathcal{E}_i, \ell_i)\}_{i \in \mathcal{I}}$ connected by surjective refinement maps with compatible charges. The Stone spaces $\{S(\mathcal{E}_i)\}$ form a projective system. The question becomes whether the charges propagate coherently to σ-additive measures throughout — and if so, what condition on the family is necessary and sufficient.

The paper establishes three results. The negative result (Section 1.3) shows that the extension question has no finitary answer even in the system setting: the structural regress is irreducible, and no chain of finite refinements resolves it. The incompleteness result (Section 1.4) shows that nontrivial refinement is an irreducibly finer operation: a Boolean algebra $\mathcal{E}$ admitting a nontrivial refinement contains events not pullable from below, and a level at which $\mathcal{E} = \sigma(\mathcal{E})$ admits no nontrivial refinement at all. The positive result (Section 1.5) identifies the correct characterisation. The natural question — whether index-layer conditions (directedness, compatibility) force σ-additivity —

turns out to be malformed: it conflates conditions on the directed system's structure with conditions on the charges it carries. We show that index-layer coherence does not force σ-additivity (the finite-cofinite counterexample), and that σ-additivity is equivalent to collective exhaustion, a purely valuation-layer condition (Theorem 1.17).

Section 2.8 locates the results and states what remains open.

1.1.1 Organisation

Section 2.2 sets up the single-algebra case: a Boolean algebra $\mathcal{E}$ with a finitely-additive charge ℓ on it, with no topology and no σ-algebra assumed.

Section 1.3 establishes the negative result: exhaustiveness is trivial, the real obstruction is continuity at $\emptyset$ for sequences escaping $\mathcal{E}$, and no finitary condition on any single level can supply this.

Section 1.4 establishes the incompleteness result: a nontrivial refinement introduces events not pullable from the coarser level, and a level with $\mathcal{E} = \sigma(\mathcal{E})$ admits no nontrivial refinement.

Section 1.5 completes the analysis: it proves the independence of index- and valuation-layer conditions and establishes the main equivalence (collective exhaustion $\iff$ σ-additive extensibility).

Section 2.8 locates the results within the Stone duality picture and states what remains open.

1.2 Single-algebra setup

We begin with a single Boolean algebra and its charge, setting aside the directed system structure to isolate the extension question.

Definition 1.1 (Discriminability structure)**.** A *discriminability structure* is a pair $(\mathcal{E}, \ell)$ where:

(i) $\mathcal{E}$ is a Boolean algebra of subsets of some set O — closed under finite union, intersection, and complementation, containing $\emptyset$ and O.

(ii) $\ell : \mathcal{E} \to [0, 1]$ is a normalized finitely-additive valuation: $\ell(\emptyset) = 0$, $\ell(O) = 1$, and $\ell(A \cup B) = \ell(A) + \ell(B)$ whenever $A \cap B = \emptyset$.

Via Stone's representation theorem, $\mathcal{E}$ is isomorphic to the clopen algebra of a compact totally-disconnected space $S(\mathcal{E})$, the Stone space of $\mathcal{E}$, and $\sigma(\mathcal{E})$ is the Baire σ-algebra of $S(\mathcal{E})$.

No σ-algebra is assumed on O. No topology is assumed.

Definition 1.2 (The gap)**.** For a discriminability structure $(\mathcal{E}, \ell)$, the *gap* is the set $\sigma(\mathcal{E}) \setminus \mathcal{E}$ — events in the σ-algebra generated by $\mathcal{E}$ that are not themselves in $\mathcal{E}$.

The gap consists of events reachable by countable Boolean operations on $\mathcal{E}$ but not contained in $\mathcal{E}$ itself.

Definition 1.3 (Extension)**.** We say ℓ *extends* to $\sigma(\mathcal{E})$ if there exists a σ-additive measure μ on $\sigma(\mathcal{E})$ with $\mu|_{\mathcal{E}} = \ell$. The extension is *unique* if any two such μ agree on all of $\sigma(\mathcal{E})$.

The central question of this paper is: what properties of $(\mathcal{E}, \ell)$ are necessary and sufficient for such an extension to exist and be unique?

1.3 The negative result: finitary conditions cannot suffice

1.3.1 Exhaustiveness is trivial

The Stone space $S(\mathcal{E})$ of a Boolean algebra $\mathcal{E}$ is compact by construction. But compactness of $S(\mathcal{E})$ does not force a given charge on $\mathcal{E}$ to extend to a Borel measure on $S(\mathcal{E})$: compactness is a property of the space, not the measure. The measure-theoretic condition is tightness — equivalently, continuity at $\emptyset$. The question is whether this can be witnessed finitely. It cannot.

The first natural candidate is exhaustiveness.

Definition 1.4 (Exhaustiveness)**.** A finitely-additive valuation $\ell : \mathcal{E} \to [0, 1]$ is *exhaustive* if $\ell(E_n) \to 0$ for every disjoint sequence $(E_n)_{n\geq 1} \subset \mathcal{E}$.

Proposition 1.5. *Every normalized finitely-additive valuation on a Boolean algebra is exhaustive.*

Proof. Let (E_n) be a disjoint sequence in $\mathcal{E}$. For any N, $\sum_{n=1}^{N} \ell(E_n) = \ell\big(\bigcup_{n=1}^{N} E_n\big) \leq \ell(O) = 1$ by finite additivity and normalization. Hence $\sum_{n=1}^{\infty} \ell(E_n) \leq 1$, so $\ell(E_n) \to 0$. $\square$

Exhaustiveness carries no information about extensibility. It is a consequence of normalization alone, independent of any coherence condition or structural hypothesis.

1.3.2 The real obstruction

The correct extension condition is continuity from above at $\emptyset$.

Definition 1.6 (Continuity at $\emptyset$)**.** A finitely-additive valuation $\ell : \mathcal{E} \to [0, 1]$ is *continuous at* $\emptyset$ if: for every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}$ with $\bigcap_{n=1}^{\infty} E_n = \emptyset$ in $\sigma(\mathcal{E})$, we have $\ell(E_n) \to 0$.

Remark 1.7. The intersection $\bigcap_n E_n$ in Definition 1.6 is taken in $\sigma(\mathcal{E})$, not in $\mathcal{E}$. The condition is therefore not verifiable from within $\mathcal{E}$ alone: confirming $\bigcap_n E_n = \emptyset$ requires access to $\sigma(\mathcal{E})$. This is precisely what the regress below captures.

Theorem 1.8 (Extension criterion). *A normalized finitely-additive $\ell : \mathcal{E} \to [0,1]$ extends to a σ-additive measure on $\sigma(\mathcal{E})$ if and only if ℓ is continuous at $\emptyset$. The extension is then unique.*

Proof. Standard: see Halmos [4] or Fremlin [3] Vol. 1. The forward direction is immediate from σ-additivity of the extension. For the converse, continuity at $\emptyset$ is precisely the condition used in the standard proof of the Carathéodory extension to pass from finite additivity to σ-additivity on the generated σ-algebra. □

1.3.3 The regress

In a directed system, one might hope that a finer level $j \geq i$ can witness the emptiness that level i cannot see, and that compatibility then forces continuity at $\emptyset$ at level i.

Proposition 1.9 (The regress). *Let $j \geq i$ in a directed system with refinement map $\pi : O_j \to O_i$. Suppose $E_1 \supseteq E_2 \supseteq \cdots$ in $\mathcal{E}_i$ with $\bigcap_n \pi^{-1}(E_n) = \emptyset$ in $\mathcal{E}_j$. Compatibility gives $\ell_i(E_n) = \ell_j(\pi^{-1}(E_n))$. But this does not force $\ell_j(\pi^{-1}(E_n)) \to 0$: the same obstruction reappears at level j, since finite additivity on $\mathcal{E}_j$ alone does not imply continuity at $\emptyset$ for sequences decreasing to an element of $\sigma(\mathcal{E}_j)$.*

The regress is structural. Pushing the witnessing one level up does not resolve it: it only relocates it. No finite chain of refinements can supply continuity at $\emptyset$ from purely finitary conditions.

Corollary 1.10. *No condition stateable purely in terms of the Boolean algebra structure of $\mathcal{E}_i$ and the finitely-additive charges $\{\ell_j\}_{j \geq i}$ at finitely many levels is necessary and sufficient for ℓ_i to extend to $\sigma(\mathcal{E}_i)$.*

Something external to any single level is required. The three candidates are: topology (compactness witnesses emptiness), a colimit algebra (a direct limit in the directed system whose event algebra contains the limit events), or an explicit continuity axiom (honest but circular). The remainder of this paper develops the colimit option, which is intrinsic to the directed system framework.

1.4 The structure of nontrivial refinement

Before turning to the positive result, we establish what nontrivial refinement actually entails at the level of the Boolean algebras.

Definition 1.11 (Nontrivial refinement). Let $\pi : O_j \to O_i$ be a surjective refinement map between levels $j \geq i$ in a directed system, inducing the Boolean algebra homomorphism $\pi^{-1} : \mathcal{E}_i \to \mathcal{E}_j$. The refinement is *nontrivial* if there exists $A \in \mathcal{E}_j$ with $A \notin \pi^{-1}(\mathcal{E}_i)$ — that is, A is not the preimage of any event at level i.

Proposition 1.12 (Nontrivial refinement introduces new events). *If $j \geq i$ is a nontrivial refinement, then $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$: the finer algebra strictly extends the pullback of the coarser.*

Proof. Immediate from the definition: nontriviality asserts the existence of $A \in \mathcal{E}_j \setminus \pi^{-1}(\mathcal{E}_i)$. □

Remark 1.13. The contrapositive is equally important: if $\mathcal{E}_j = \pi^{-1}(\mathcal{E}_i)$ for every $j \geq i$, then every finer level distinguishes exactly the events pullable from level i — the refinement order above i is algebraically trivial. Nontriviality of refinement is therefore the condition that the directed system is genuinely increasing in discriminative power.

Remark 1.14 (Relation to the gap). The gap $\sigma(\mathcal{E}_i) \setminus \mathcal{E}_i$ is a separate condition: it concerns events reachable by countable operations on $\mathcal{E}_i$ that are not themselves in $\mathcal{E}_i$. When $\mathcal{E}_i$ is not already a σ-algebra — in particular when O_i is uncountable and $\mathcal{E}_i$ does not contain all subsets of O_i — this gap is nonempty. Under Stone duality, the gap corresponds to the Baire sets of $S(\mathcal{E}_i)$ that are not clopen: a level with $\mathcal{E}_i = \sigma(\mathcal{E}_i)$ would correspond to a Stone space whose entire Baire σ-algebra consists of clopens, a highly degenerate condition. In the settings of interest — outcome spaces that are not finite discrete — the gap is nonempty, and the extension question is nontrivial.

1.5 Collective exhaustion and the characterisation of σ-additivity

The negative result established that no finitary or index-layer condition can force σ-additivity. This section completes the picture: we identify the exact valuation-layer condition that characterises σ-additive extensibility, show it is independent of the index-layer conditions, and prove the equivalence.

Proposition 1.15 (Independence of index and valuation layers). *Sequential upper-directedness together with compatibility of finitely-additive charges does not force σ-additivity. There exists a compatible family of normalized finitely-additive charges on a sequentially upper-directed directed system that fails σ-additive extension at every level.*

Proof. Let $\iota = \mathbb{N} \cup \{\omega\}$ with $n \leq \omega$ for all $n \in \mathbb{N}$, directed by this order. This is sequentially upper-directed: ω is a universal upper bound for every sequence. Take all outcome spaces to be $\mathbb{Q}$, all event algebras to be the finite-cofinite algebra $\mathcal{E} = \{E \subseteq \mathbb{Q} :$ E is finite or E^c is finite$\}$, and all refinement maps to be the identity. Define $\ell(E) = 0$ if E is finite and $\ell(E) = 1$ if E is cofinite. This is a normalized finitely-additive content, and compatibility holds trivially.

Fix an enumeration $q : \mathbb{N} \to \mathbb{Q}$. Let $F_n = \{q(0), \ldots, q(n)\}$. Each F_n is finite, so $\ell(F_n) = 0$ for all n. Since q is surjective, $\bigcup_n F_n = \mathbb{Q}$, and $\ell(\mathbb{Q}) = 1$. σ-subadditivity would give $1 \leq \sum_n \ell(F_n) = 0$: contradiction. So ℓ fails σ-additive extension at every level. □

The counterexample establishes that index-layer coherence is insufficient: the σ-algebra is not derivable from structural conditions on the directed system alone. The charge is perfectly coherent within the finite-cofinite algebra; the failure is a valuation-layer pathology invisible to the index structure.

Definition 1.16 (Collectively exhaustive)**.** A compatible family $\{\ell_i\}$ of normalized finitely-additive charges on a directed system is *collectively exhaustive* if: for every index $i \in \mathcal{I}$ and every decreasing sequence $E_1 \supseteq E_2 \supseteq \cdots$ of events in $\mathcal{E}_i$ with $\bigcap_n E_n = \emptyset$, there exists $j \geq i$ such that $\ell_j(\pi_{ji}^{-1}(E_n)) \to 0$.

Collective exhaustion is a *valuation-layer* condition: it places a requirement on how the charges behave, not on the index structure of the directed system. It says that no level can assign persistent mass to events that the system as a whole sees as empty. The counterexample above fails collective exhaustion: for the sequence $E_n = \mathbb{Q} \setminus F_n$ (cofinite sets shrinking to $\emptyset$), every level assigns charge 1 to $\pi_{ji}^{-1}(E_n) = E_n$, and no witness exists anywhere in the system.

Theorem 1.17 (Collective exhaustion characterises σ-additivity)**.** *Let $\{\ell_i\}$ be a normalized compatible family of finitely-additive charges on a directed system indexed by $\mathcal{I}$. The following are equivalent:*

(i) The family is collectively exhaustive.

(ii) ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$ for every $i \in \mathcal{I}$.

Proof. *(i) $\Rightarrow$ (ii).* Fix $i \in \mathcal{I}$ and a decreasing sequence $E_n \searrow \emptyset$ in $\mathcal{E}_i$. By collective exhaustion, find $j \geq i$ with $\ell_j(\pi_{ji}^{-1}(E_n)) \to 0$. By compatibility, $\ell_i(E_n) = \ell_j(\pi_{ji}^{-1}(E_n)) \to 0$. So ℓ_i is continuous at $\emptyset$. By the extension criterion (Theorem 1.8), ℓ_i extends uniquely to a σ-additive measure on $\sigma(\mathcal{E}_i)$.

(ii) $\Rightarrow$ (i). Suppose ℓ_i extends to μ_i for every $i \in \mathcal{I}$. Let $E_n \searrow \emptyset$ in $\mathcal{E}_i$. Then $\ell_i(E_n) = \mu_i(E_n) \to \mu_i(\emptyset) = 0$ by σ-additivity of μ_i and normalization (which ensures $\mu_i(E_0) \leq \mu_i(O_i) = 1 < \infty$, the finiteness condition for continuity from above). Take $j = i$. $\square$

Remark 1.18 (The dissolution). Proposition 1.15 shows that index-layer conditions (sequential upper-directedness, compatibility) do not force σ-additivity. These layers are independent: the index layer (the preorder, refinement maps, sequential upper-directedness) and the valuation layer (the charges, compatibility, continuity at $\emptyset$) carry different information.

The correct question is: what valuation-layer condition characterises σ-additive extensibility? Theorem 1.17 answers it: collective exhaustion. It is not an axiom imposed from outside — it is the exact condition separating directed systems in which every convergence is witnessed somewhere in the refinement order from those in which it is not.

Sequential upper-directedness plays no role in Theorem 1.17: it is a per-level result. Sequential upper-directedness enters when one asks whether the per-level extensions assemble into a global probability space over Ω — that is the next question.

1.6 Discussion

Under Stone duality, $\mathcal{E}_i$ corresponds to the clopen algebra of a compact totally-disconnected space and $\sigma(\mathcal{E}_i)$ to its Baire σ-algebra; the extension question is whether a charge on the clopens extends to a Borel measure. The present paper answers this for directed systems without topological hypotheses on the outcome spaces: collective exhaustion is the necessary and sufficient condition, playing the role that tightness plays on the Stone space.

Theorem 1.17 is a per-level result. Sequential upper-directedness plays no role in it: that condition enters when asking whether the per-level extensions $\{\mu_i\}$ assemble into a single σ-additive measure on the projective limit $\Omega = \varprojlim O_i$. For compact outcome spaces this follows from the Prokhorov extension theorem; for standard Borel spaces from Musiał's theorem via perfectness. The topology-free case remains open.

Proposition 1.12 establishes that nontrivial refinement strictly extends the pullback algebra: $\pi^{-1}(\mathcal{E}_i) \subsetneq \mathcal{E}_j$. The gap $\sigma(\mathcal{E}_i) \setminus \mathcal{E}_i$ is nonempty whenever $\mathcal{E}_i$ is not already a σ-algebra (Remark 1.14). Constructing a specific element of this gap from the refinement data — connecting to the Stone space picture where the gap corresponds to Baire sets not clopen in $S(\mathcal{E}_i)$ — is the concrete content of the incompleteness claim.

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From Discriminability to Probability: The First Transition

Chapter 1 establishes the structural conditions under which a charge on a Boolean algebra extends to a σ-additive measure. The answer — collective exhaustion — is a condition on the family of charges, not on the Boolean algebras themselves. This interstitial draws out the conceptual significance of that result before turning to the global extension question of Chapter 2.

The gap as horizon. The gap $\sigma(\mathcal{E}_Q) \setminus \mathcal{E}_Q$ consists of events reachable by countable Boolean operations on $\mathcal{E}_Q$ but not themselves in $\mathcal{E}_Q$. The incompleteness theorem shows this gap is not a deficiency: any query admitting genuine refinement must have a nonempty gap. Under Stone duality, $\mathcal{E}_Q$ corresponds to the clopen algebra of the Stone space $S(\mathcal{E}_Q)$, and $\sigma(\mathcal{E}_Q)$ to its Baire σ-algebra. The gap is the collection of Baire sets that are not clopen — the limit structure of the Stone space that lies beyond the clopen topology. A query with no gap would correspond to a Stone space whose Baire σ-algebra coincides with the clopen algebra: a highly degenerate space admitting no interesting limit structure. The gap is therefore the space into which refinements reach; its presence is what makes refinement meaningful.

The σ-algebra as forced structure. The extension criterion (Theorem 2.3 of Chapter 1) says that $\sigma(\mathcal{E}_Q)$ enters the picture not as a postulate but as the minimal structure that must be present for ℓ_Q to be extendable at all. A charge that extends to $\sigma(\mathcal{E}_Q)$ is one that already "knows about" the gap — it assigns vanishing weight to sequences converging to events it cannot directly represent. The σ-algebra is the closure forced by that knowledge.

Collective exhaustion and the question of origin. The main theorem of Chapter 1 establishes that collective exhaustion is precisely the valuation-layer condition equivalent to σ-additive extensibility. The result separates two questions that are easily conflated: whether the *structural* conditions on a query system (directedness, compatibility) force σ-additivity, and whether the *valuations* carried by the system do. The former is impossible (the finite-cofinite counterexample); the latter is characterised exactly (collective exhaustion). The σ-algebra is not derivable from the index structure of the query system alone, nor is it a free assumption: it is the content of a specific and checkable condition on the family of charges.

One can ask, in a more philosophical register, what it means for a system to satisfy collective exhaustion. The condition says that no level can persistently assign mass to events that vanish globally: every convergence in the system is witnessed at some finer level. In this sense collective exhaustion is the mathematical expression of the idea that the system *models a coherent world* — one in which events that become empty do so in a way that is visible from within the system. A system failing collective exhaustion contains an observer who permanently assigns positive probability to events the system as a whole regards as impossible. The σ-algebra, then, is the structure that a coherent world forces on the local event languages of its observers.

The projective limit $\Omega = \varprojlim O_Q$ plays a role analogous to what Kant called a *regulative ideal*: it is not itself one of the queries in the system, and no single observer occupies it, but it is the standard against which the coherence of the system is measured. Continuity at $\emptyset$ for ℓ_Q is the local expression of a global orientation toward Ω: the charge at level Q must behave as if Ω exists, even though Q cannot see it directly. Collective exhaustion makes this requirement non-vacuous: it is not a private act of faith by each individual observer but a condition verifiable from within the system.

From per-level extensions to a global measure. Chapter 1 gives per-level extensions μ_Q on $\sigma(\mathcal{E}_Q)$ for each Q. The next question is whether these assemble into a single σ-additive measure on Ω. This is strictly harder: it requires the local extensions to concentrate mass on Ω rather than dispersing it into the ambient product $\prod_Q O_Q$. Sequential upper-directedness — the condition that every sequence of queries has a common refinement — is the structural hypothesis that makes global assembly possible. It plays no role in the per-level result but is essential for the projective limit.

The passage from Chapter 1 to Chapter 2 is therefore the passage from local to global: from the characterisation of σ-additivity at each level to the construction of a single probability measure on the world the system describes.

Chapter 2

Finitely Additive Observable Laws and the Prokhorov Extension

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Abstract

The observational foundations program constructs a canonical probability space from a compatible family of observable laws, taking countable additivity of those laws as a hypothesis. This paper asks when that hypothesis can itself be derived. We show that for a countable Hausdorff query system with continuous refinement maps, any finitely compatible family of finitely additive normalized valuations satisfying a *surjective projective uniform tightness* condition admits a unique extension to a σ-additive probability measure on the canonical realization space. Countable additivity is not assumed locally — it is forced globally by the combination of projective compactness and sequential refinement structure.

The result is a Prokhorov-type extension theorem in the language of query systems. It complements the main extension theorem of *Observational Foundations of Probability* (which takes σ-additive marginals as input) by identifying conditions under which those marginals need not be assumed: they emerge from a tightness condition on the compatible family as a whole. The surjective projective uniform tightness condition is the query-system analogue of Prokhorov tightness, with the surjectivity of bonding maps on compact cores playing the role that single-space compactness plays in the classical theory.

2.1 Introduction

The foundational question. The observational program constructs probability from observation. Its central extension theorem (*Observational Foundations of Probability*, hereafter Paper 1) shows that a compatible family of observable laws $\{\nu_Q\}$ on a query system $(\mathcal{Q}, \preceq)$ determines a unique probability measure P on the canonical realization space Ω. The construction is a representation theorem: given that each query Q carries a countably additive law ν_Q, the global P is assembled from those laws.

The question this paper asks is foundationally prior:

> *Must one assume countable additivity at each query level, or can it be derived from the structure of the compatible family?*

This is Stopping Point 2 of the program: the assumption that each ν_Q is σ-additive is the fuel on which the extension theorem runs. The present paper identifies conditions under which that fuel can itself be derived — conditions that are properties of the *system* rather than of individual marginals.

The answer. We show that under a condition called *surjective projective uniform tightness* (SPUT), starting from only finitely additive normalized valuations $\{\ell_Q\}$ satisfying finite compatibility, one can derive a σ-additive global extension. The argument has two steps.

The first step is analytic: SPUT supplies compatible compact cores $\{K_Q\}$ whose inverse limit $K_\Omega \subseteq \Omega$ is compact and nonempty. Any decreasing sequence of cylinder sets with uniformly positive mass must meet K_Ω at each level; since K_Ω is compact, the intersection is nonempty, contradicting the assumption that the sequence decreases to $\emptyset$. This forces the cylinder premeasure μ_0 to be σ-subadditive, hence σ-additive by finite additivity and Carathéodory extension.

The second step is geometric: the extended measure lives on Ω rather than the ambient product space. A graph-support lemma shows that for each comparable pair $Q_1 \preceq Q_2$, the joint law of $(\mathrm{eval}_{Q_1}, \mathrm{eval}_{Q_2})$ under P is supported on the graph of $\pi^{Q_2}_{Q_1}$. For countable $\mathcal{Q}$, a union over countably many null defect sets gives $P(\Omega) = 1$.

What SPUT is. SPUT asks that for every $\varepsilon > 0$ one can find compact sets $K_Q \subset O_Q$ at each query level such that: (i) the bonding maps restrict to surjections between compact sets, $\pi^{Q_2}_{Q_1}(K_{Q_2}) = K_{Q_1}$; and (ii) the mass outside the compact core is uniformly small, $\ell_Q(O_Q \setminus K_Q) < \varepsilon$ for every Q. Condition (i) — surjectivity of the bonding maps on the compact cores — is the critical one. It ensures the compact inverse limit $K_\Omega = \varprojlim K_Q$ is nonempty with surjective projections, which is exactly what the compactness argument requires. Without surjectivity, K_Ω can be empty even when each K_Q is nonempty.

Comparison to classical Prokhorov. In classical Prokhorov tightness [10], one controls mass concentration in a single space: for every $\varepsilon > 0$, compact $K \subset X$ with $\mu_n(X \setminus K) < \varepsilon$. The query-system analogue must control concentration *projectively*, across all query levels simultaneously, and must ensure the compact sets assemble coherently in the inverse limit. SPUT is exactly this coherent projective tightness condition, with surjectivity of bonding maps playing the role that single-space compactness plays classically [9].

Relation to Paper 1. The two extension theorems are complementary:

	Paper 1 (realizability route)	This paper (tightness route)
Input measures	σ-additive ν_Q	finitely additive ℓ_Q
Key condition	realizability of Ω	surjective projective uniform tightness
Topology required	no	Hausdorff O_Q, continuous π
Conclusion	σ-additive P on Ω	σ-additive P on Ω

Paper 1 takes σ-additivity as given and derives a global measure from realizability. This paper takes only finite additivity and derives both σ-additivity and realizability from tightness. The price is topological structure on the query system.

Structure of the paper. Section 2.2 fixes notation and introduces finitely additive compatible families on query systems. Section 2.3 defines SPUT, motivates the surjectivity condition, and gives examples. Section 2.4 proves σ-subadditivity on closed cylinder sequences (the intermediate theorem). Section 2.5 extends to general Borel cylinders using Alexandrov regularity. Section 2.6 establishes concentration on Ω via the graph-support lemma. Section 2.7 assembles the full theorem and compares it to Paper 1. Section 2.8 discusses what the result achieves for the foundational program, what it does not yet achieve, and the open frontier.

2.2 Setup

Definition 2.1 (Query system)**.** A *query system* is a tuple $(\mathcal{Q}, \preceq, \{O_Q\}, \{\pi_{Q_1}^{Q_2}\})$ where:

- $(\mathcal{Q}, \preceq)$ is a countable directed preorder (the *query index set*);
- each $Q \in \mathcal{Q}$ is assigned a measurable outcome space $(O_Q, \mathcal{B}(O_Q))$;
- for each $Q_1 \preceq Q_2$, there is a measurable map $\pi_{Q_1}^{Q_2} : O_{Q_2} \to O_{Q_1}$ (the *refinement map*) satisfying $\pi_{Q_1}^{Q_1} = \mathrm{id}$ and $\pi_{Q_1}^{Q_3} = \pi_{Q_1}^{Q_2} \circ \pi_{Q_2}^{Q_3}$ whenever $Q_1 \preceq Q_2 \preceq Q_3$.

The system is *sequentially upper-directed* if for every countable family $\{Q_n\}_{n\geq 1} \subset \mathcal{Q}$ there exists $Q_* \in \mathcal{Q}$ with $Q_* \succeq Q_n$ for all n.

Remark 2.2 (Topological enrichment). Throughout this paper the outcome spaces O_Q are assumed to be Hausdorff topological spaces with $\mathcal{B}(O_Q)$ the Borel σ-algebra, and the refinement maps $\pi_{Q_1}^{Q_2}$ are assumed to be continuous. These topological conditions are not required by Paper 1 and are the price paid for starting from only finitely additive marginals.

Definition 2.3 (Canonical realization space)**.** The *canonical realization space* is the projective limit

$$\Omega := \varprojlim_{Q \in \mathcal{Q}} O_Q = \left\{ \omega \in \prod_{Q \in \mathcal{Q}} O_Q \;\middle|\; \pi_{Q_1}^{Q_2}(\omega_{Q_2}) = \omega_{Q_1} \;\forall\, Q_1 \preceq Q_2 \right\},$$

equipped with evaluation maps $\mathrm{eval}_Q : \Omega \to O_Q$, $\omega \mapsto \omega_Q$.

Since each $\pi_{Q_1}^{Q_2}$ is continuous and each O_Q is Hausdorff, each coherence constraint set $\{\omega : \pi_{Q_1}^{Q_2}(\omega_{Q_2}) = \omega_{Q_1}\}$ is closed in the product. Since $\mathcal{Q}$ is countable, Ω is a countable intersection of closed sets, hence closed in $\prod_Q O_Q$.

The *cylinder algebra* $\mathcal{C}$ is generated by sets $\mathrm{Cyl}(Q, A) := \mathrm{eval}_Q^{-1}(A)$ for $Q \in \mathcal{Q}$, $A \in \mathcal{B}(O_Q)$, and the *observable σ-field* is $\sigma(\mathcal{Q}) := \sigma(\mathcal{C})$.

Definition 2.4 (Finitely compatible family)**.** A family $\{\ell_Q\}_{Q\in\mathcal{Q}}$ of normalized valuations $\ell_Q : \mathcal{B}(O_Q) \to [0,1]$ (with $\ell_Q(O_Q) = 1$) is *finitely compatible* if for every $Q_1 \preceq Q_2$,

$$(\pi^{Q_2}_{Q_1})_{\#}\ell_{Q_2} = \ell_{Q_1},$$

i.e. $\ell_{Q_1}(A) = \ell_{Q_2}((\pi^{Q_2}_{Q_1})^{-1}(A))$ for all $A \in \mathcal{B}(O_{Q_1})$. Each ℓ_Q is assumed to be finitely additive (but not necessarily σ-additive).

Remark 2.5 (The cylinder premeasure). Given a finitely compatible family $\{\ell_Q\}$, define the *cylinder premeasure* $\mu_0 : \mathcal{C} \to [0,1]$ by

$$\mu_0(\mathrm{Cyl}(Q, A)) := \ell_Q(A).$$

Finite compatibility ensures μ_0 is well-defined (independent of which query level is used to represent a given cylinder) and finitely additive on $\mathcal{C}$. The central question is when μ_0 is σ-additive, so that Carathéodory's theorem [2, 3] extends it to a probability measure on $\sigma(\mathcal{Q})$.

2.3 Surjective projective uniform tightness

Definition 2.6 (Surjective projective uniform tightness)**.** A finitely compatible family $\{\ell_Q\}$ is *surjectively projectively uniformly tight* (SPUT) if for every $\varepsilon > 0$ there exists a system $\{K_Q\}_{Q\in\mathcal{Q}}$ of compact sets $K_Q \subset O_Q$ such that:

(i) *Surjective bonding:* $\pi^{Q_2}_{Q_1}(K_{Q_2}) = K_{Q_1}$ for all $Q_1 \preceq Q_2$;

(ii) *Uniform mass control:* $\ell_Q(O_Q \setminus K_Q) < \varepsilon$ for every $Q \in \mathcal{Q}$.

The two conditions play distinct roles. Condition (ii) ensures that the compact cores capture most of the mass at every query level. Condition (i) ensures the compact cores assemble coherently across levels, so their inverse limit is nonempty.

Remark 2.7 (Why surjectivity is necessary). If condition (i) required only $\pi^{Q_2}_{Q_1}(K_{Q_2}) \subseteq K_{Q_1}$ (containment rather than equality), the inverse limit $K_\Omega := \Omega \cap \prod_Q K_Q$ could be empty even when each K_Q is nonempty. The classical example: let $O_Q = [0,1]$, $K_Q = [0, 1 - 1/Q]$, with refinement maps the identity. Then each K_Q is nonempty and the bonding maps satisfy containment, but $\bigcap_Q K_Q = \emptyset$. Surjectivity — $\pi(K_{Q_2}) = K_{Q_1}$ exactly — rules out such "shrinking core" phenomena and guarantees $K_\Omega \neq \emptyset$ with surjective projections (Proposition 2.13).

Remark 2.8 (Comparison to classical Prokhorov tightness). Classical tightness for a family of measures $\{\mu_n\}$ on a single metric space X asks: for every $\varepsilon > 0$, a compact $K \subset X$ with $\mu_n(K) \geq 1 - \varepsilon$ for all n. SPUT is the projective analogue: compact sets must be chosen at every query level simultaneously, and they must assemble into a compact subset of Ω via the surjectivity condition. The surjectivity of bonding maps is the multi-space analogue of the fact that K is a single space (automatically surjective over itself) in the classical case.

Proposition 2.9 (Compact outcome spaces imply SPUT)**.** *Suppose each outcome space O_Q is compact and each refinement map $\pi^{Q_2}_{Q_1}$ is surjective. Then every finitely compatible family $\{\ell_Q\}$ of normalized valuations is SPUT.*

Proof. For any $\varepsilon > 0$, take $K_Q := O_Q$ for every $Q \in \mathcal{Q}$. Then:

(i) $\pi^{Q_2}_{Q_1}(K_{Q_2}) = \pi^{Q_2}_{Q_1}(O_{Q_2}) = O_{Q_1} = K_{Q_1}$ for all $Q_1 \preceq Q_2$, by surjectivity of the refinement maps.

(ii) $\ell_Q(O_Q \setminus K_Q) = \ell_Q(\emptyset) = 0 < \varepsilon$ for every Q.

Both SPUT conditions are satisfied with the bound being 0, not merely $< \varepsilon$. □

Remark 2.10 (What this means)*.* When the outcome spaces are compact, the SPUT condition is not a constraint on the valuations $\{\ell_Q\}$ at all — it is a property of the query system architecture. Any finitely compatible family, however pathological, satisfies SPUT because the compact cores are simply the full spaces. The surjectivity of refinement maps then supplies condition (i) for free. No additional tightness assumption on the observer's laws is required.

Remark 2.11 (Surjective maps without compact spaces)*.* When the outcome spaces are not compact, surjectivity of the refinement maps alone does not imply SPUT. But it does simplify the construction: given classical tightness of each ℓ_Q (compact $K_Q \subset O_Q$ with $\ell_Q(K_Q) \geq 1 - \varepsilon$), one can build compatible compact cores by setting $K'_Q := \pi^{Q_*}_{Q}(K_{Q_*})$ for a sufficiently fine Q_*. Surjectivity ensures $\pi^{Q_2}_{Q_1}(K'_{Q_2}) = K'_{Q_1}$, and the mass outside the core is controlled by the tightness of ℓ_{Q_*}. Thus the delay query systems of *Observational Probability from Delay Queries* (where refinement maps are surjective coordinate projections) satisfy SPUT whenever the marginals are individually tight.

Definition 2.12 (Compact core)**.** Given a SPUT system $\{K_Q\}$, the *compact core* of Ω is

$$K_\Omega := \Omega \cap \prod_{Q \in \mathcal{Q}} K_Q = \{\omega \in \Omega \mid \omega_Q \in K_Q \ \forall\, Q \in \mathcal{Q}\}.$$

Proposition 2.13 (Properties of the compact core)**.** *Under SPUT:*

(i) K_Ω is compact.

(ii) K_Ω is nonempty, and $\mathrm{eval}_Q(K_\Omega) = K_Q$ *for every $Q \in \mathcal{Q}$.*

Proof. (i) The product $\prod_Q K_Q$ is compact by Tychonoff's theorem [4]. Since Ω is closed in $\prod_Q O_Q$ (as each coherence constraint is a closed condition), $K_\Omega = \Omega \cap \prod_Q K_Q$ is a closed subset of a compact space, hence compact.

(ii) The system $(K_Q, \pi^{Q_2}_{Q_1}|_{K_{Q_2}})$ is an inverse system of nonempty compact Hausdorff spaces. By condition (i) of SPUT, the bonding maps $\pi^{Q_2}_{Q_1}|_{K_{Q_2}} : K_{Q_2} \to K_{Q_1}$ are surjective. By the *inverse limit theorem* [4, 6] for compact Hausdorff spaces with surjective bonding maps over a directed index set, $K_\Omega = \varprojlim K_Q$ is nonempty and each projection $\mathrm{eval}_Q : K_\Omega \to K_Q$ is surjective. □

Remark 2.14 (The inverse limit theorem). Proposition 2.13(ii) uses the classical result that an inverse system of nonempty compact Hausdorff spaces with surjective bonding maps has a nonempty inverse limit with surjective projections. This follows by expressing K_Ω as an intersection in the compact product $\prod_Q K_Q$, where each finite sub-intersection is nonempty by surjectivity, and applying the finite intersection property. The directed-set version of this argument is standard; it has been formalized in Lean 4 as `compactInverseLimit_nonempty`, using upper-directedness to supply a common upper bound at each finite stage and proof irrelevance on the partial-order witnesses to close the coherence conditions.

2.4 The intermediate theorem: closed cylinders

The key analytic step is to show that the cylinder premeasure μ_0 is σ-subadditive on sequences of closed cylinders.

Theorem 2.15 (Closed cylinder σ-subadditivity). *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system with Hausdorff outcome spaces and continuous refinement maps. Let $\{\ell_Q\}$ be a finitely compatible SPUT family. If $C_N = \mathrm{Cyl}(Q_N, F_N) \downarrow \emptyset$ is a decreasing sequence of cylinder sets with each $F_N \subseteq O_{Q_N}$ closed, then*

$$\mu_0(C_N) \to 0.$$

Proof. Suppose for contradiction that $\mu_0(C_N) \geq \varepsilon > 0$ for all N.

Step 1 (Compact core). Apply SPUT with parameter $\varepsilon/2$: obtain compact sets $\{K_Q\}$ with surjective bonding maps and $\ell_Q(O_Q \setminus K_Q) < \varepsilon/2$ for all Q.

Step 2 (K_Ω is compact and nonempty with surjective projections). By Proposition 2.13, K_Ω is compact and $\mathrm{eval}_{Q_N}(K_\Omega) = K_{Q_N}$ for every N.

Step 3 (Each $C_N \cap K_\Omega$ is nonempty). Since $\ell_{Q_N}(F_N) = \mu_0(C_N) \geq \varepsilon$ and $\ell_{Q_N}(K_{Q_N}) \geq 1 - \varepsilon/2$, we have

$$\ell_{Q_N}(F_N \cap K_{Q_N}) \geq \varepsilon - \varepsilon/2 = \varepsilon/2 > 0.$$

In particular $F_N \cap K_{Q_N} \neq \emptyset$. Since $\mathrm{eval}_{Q_N} : K_\Omega \to K_{Q_N}$ is surjective, there exists $\omega \in K_\Omega$ with $\mathrm{eval}_{Q_N}(\omega) \in F_N \cap K_{Q_N}$, so $\omega \in C_N \cap K_\Omega$.

Step 4 (Each $C_N \cap K_\Omega$ is closed in K_Ω). $C_N = \mathrm{eval}_{Q_N}^{-1}(F_N)$ where F_N is closed and eval_{Q_N} is continuous, so C_N is closed in Ω and $C_N \cap K_\Omega$ is closed in K_Ω.

Step 5 (Contradiction by compactness). The sets $\{C_N \cap K_\Omega\}$ form a decreasing sequence of nonempty closed subsets of the compact space K_Ω. By the finite intersection property,

$$\bigcap_N (C_N \cap K_\Omega) \neq \emptyset.$$

But $\bigcap_N C_N = \emptyset$ (since $C_N \downarrow \emptyset$), so $\bigcap_N (C_N \cap K_\Omega) \subseteq \bigcap_N C_N = \emptyset$. Contradiction. □

2.5 Extension to Borel cylinders

The intermediate theorem covers closed cylinders. We now extend to all Borel cylinders. The key observation is that no new hypotheses are needed: inner regularity on each compact core is automatic.

Lemma 2.16 (Alexandrov regularity is free)**.** *Let K be a compact Hausdorff space and ν a finite Borel measure on K. Then ν is inner regular: for every Borel $A \subseteq K$ and $\eta > 0$, there exists closed $F \subseteq A$ with $\nu(A \setminus F) < \eta$.*

This is Alexandrov's theorem [1, 3]; it requires only that K is compact Hausdorff, which follows from SPUT.

Theorem 2.17 (Borel cylinder σ-subadditivity)**.** *Under the hypotheses of Theorem 2.15, if $C_N = \mathrm{Cyl}(Q_N, A_N) \downarrow \emptyset$ is a decreasing sequence of Borel cylinder sets, then $\mu_0(C_N) \to 0$.*

Proof. Suppose $\mu_0(C_N) \geq \varepsilon > 0$ for all N. Set $\varepsilon' := \varepsilon/4$.

Apply SPUT with parameter ε', obtaining compact cores $\{K_Q\}$ with $\ell_Q(O_Q \setminus K_Q) < \varepsilon'$ for all Q.

For each N, apply Alexandrov regularity (Lemma 2.16) to $\ell_{Q_N}|_{K_{Q_N}}$ with tolerance $\varepsilon'/2^N$: find closed $F_N \subseteq A_N \cap K_{Q_N}$ (closed in O_{Q_N}, since K_{Q_N} is compact in the Hausdorff space O_{Q_N}) with

$$\ell_{Q_N}(A_N \cap K_{Q_N}) - \ell_{Q_N}(F_N) < \varepsilon'/2^N.$$

Define closed cylinders $D_N := \mathrm{Cyl}(Q_N, F_N)$. Then

$$\mu_0(D_N) = \ell_{Q_N}(F_N) \geq \ell_{Q_N}(A_N) - \varepsilon' - \varepsilon'/2^N \geq \varepsilon - 2\varepsilon' = \varepsilon/2 > 0.$$

Let $E_N := \bigcap_{k=1}^N D_k$. Since the system is sequentially upper-directed, finite intersections of closed cylinders are again closed cylinders (each pair (D_i, D_j) has a common refinement $Q_{ij} \succeq Q_i, Q_j$, giving $D_i \cap D_j = \mathrm{Cyl}(Q_{ij}, (\pi^{Q_{ij}}_{Q_i})^{-1}(F_i) \cap (\pi^{Q_{ij}}_{Q_j})^{-1}(F_j))$, which is a closed cylinder). So each E_N is a closed cylinder.

Moreover, since C_N is decreasing and $D_k \subseteq C_k$:

$$\mu_0(E_N) \geq \mu_0(C_N) - \sum_{k=1}^N \mu_0(C_k \setminus D_k) \geq \varepsilon - \sum_{k=1}^\infty 2\varepsilon'/2^k = \varepsilon - 2\varepsilon'(1+1) = \varepsilon/2 > 0.$$

Since $E_N \subseteq C_N$ and $C_N \downarrow \emptyset$, we have $E_N \downarrow \emptyset$ with each $\mu_0(E_N) \geq \varepsilon/2 > 0$, contradicting Theorem 2.15. □

Corollary 2.18 (σ-additivity of μ_0)**.** *Under the above hypotheses, the cylinder premeasure μ_0 is σ-additive on $\mathcal{C}$. By Carathéodory's extension theorem, μ_0 extends uniquely to a σ-additive probability measure on $\sigma(\mathcal{Q})$ over the ambient product $\prod_Q O_Q$.*

2.6 Concentration on Ω

The Carathéodory extension lives on the ambient product $\prod_Q O_Q$. We must show it concentrates on the projective limit Ω.

Lemma 2.19 (Graph support)**.** *Let P be the σ-additive extension of μ_0 to $\prod_Q O_Q$. For each $Q_1 \preceq Q_2$, the joint law of $(\mathrm{eval}_{Q_1}, \mathrm{eval}_{Q_2})$ under P is supported on the graph*

$$G_{Q_1,Q_2} := \left\{(a,b) \in O_{Q_1} \times O_{Q_2} \;\middle|\; a = \pi_{Q_1}^{Q_2}(b)\right\}.$$

Proof. It suffices to verify on measurable rectangles, which generate $\mathcal{B}(O_{Q_1}) \otimes \mathcal{B}(O_{Q_2})$. For $A \in \mathcal{B}(O_{Q_1})$ and $B \in \mathcal{B}(O_{Q_2})$:

$$\begin{aligned} P\big(\mathrm{eval}_{Q_1}^{-1}(A) \cap \mathrm{eval}_{Q_2}^{-1}(B)\big) &= \mu_0\big(\mathrm{Cyl}(Q_2, (\pi_{Q_1}^{Q_2})^{-1}(A) \cap B)\big) \\ &= \ell_{Q_2}\big((\pi_{Q_1}^{Q_2})^{-1}(A) \cap B\big), \end{aligned}$$

which equals $(\mathrm{graph}_\pi)_\# \ell_{Q_2}(A \times B)$ where $\mathrm{graph}_\pi : b \mapsto (\pi_{Q_1}^{Q_2}(b), b)$. The measure $(\mathrm{graph}_\pi)_\# \ell_{Q_2}$ is entirely supported on G_{Q_1,Q_2}.

The graph G_{Q_1,Q_2} is Borel: it is the preimage of the diagonal $\Delta_{O_{Q_1}}$ (closed in Hausdorff $O_{Q_1} \times O_{Q_1}$) under the continuous map $(a,b) \mapsto (a, \pi_{Q_1}^{Q_2}(b))$. Hence G_{Q_1,Q_2} is closed, so the statement is well-posed. $\square$

Corollary 2.20. *For each $Q_1 \preceq Q_2$, defining the defect set $D_{Q_1,Q_2} := \{\omega : \omega_{Q_1} \neq \pi_{Q_1}^{Q_2}(\omega_{Q_2})\}$, we have $P(D_{Q_1,Q_2}) = 0$.*

Theorem 2.21 (Concentration on Ω)**.** *$P(\Omega) = 1$.*

Proof. The complement decomposes as $\prod_Q O_Q \setminus \Omega = \bigcup_{Q_1 \preceq Q_2} D_{Q_1,Q_2}$. Since $\mathcal{Q}$ is countable, the preorder has countably many comparable pairs. By Corollary 2.20, each D_{Q_1,Q_2} is a P-null set. A countable union of null sets is null, so $P(\prod_Q O_Q \setminus \Omega) = 0$. $\square$

2.7 The main theorem

Theorem 2.22 (Prokhorov extension for query systems)**.** *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system in which each outcome space O_Q is a Hausdorff topological space, $\mathcal{B}(O_Q)$ is its Borel σ-algebra, and each refinement map $\pi_{Q_1}^{Q_2} : O_{Q_2} \to O_{Q_1}$ is continuous. Let $\{\ell_Q\}_{Q \in \mathcal{Q}}$ be a finitely compatible family of finitely additive normalized valuations on $\{(O_Q, \mathcal{B}(O_Q))\}$.*

If $\{\ell_Q\}$ is surjectively projectively uniformly tight (Definition 2.6), then there exists a unique σ-additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ satisfying

$$(\mathrm{eval}_Q)_\# P = \ell_Q$$

for every $Q \in \mathcal{Q}$. In particular, each ℓ_Q is σ-additive.

Proof. σ-additivity of μ_0. By Theorem 2.17, μ_0 satisfies continuity from above at $\emptyset$ on all Borel cylinders. Since μ_0 is already finitely additive, it is σ-additive on $\mathcal{C}$.

Carathéodory extension. By Carathéodory's theorem, μ_0 extends uniquely to a σ-additive probability measure, still called P, on $\sigma(\mathcal{C})$ over $\prod_Q O_Q$.

Concentration. By Theorem 2.21, $P(\Omega) = 1$. The restriction $P|_\Omega$ is the required measure on $(\Omega, \sigma(\mathcal{Q}))$.

Marginal recovery. For any $Q \in \mathcal{Q}$ and $A \in \mathcal{B}(O_Q)$: $P(\mathrm{eval}_Q^{-1}(A)) = \mu_0(\mathrm{Cyl}(Q, A)) = \ell_Q(A)$.

Uniqueness. The observable σ-field $\sigma(\mathcal{Q})$ is generated by the cylinder algebra $\mathcal{C}$. Any two σ-additive probability measures on $(\Omega, \sigma(\mathcal{Q}))$ with the same ℓ_Q-marginals agree on $\mathcal{C}$ and hence on $\sigma(\mathcal{Q})$. □

Remark 2.23 (Induced σ-additivity). The last sentence of Theorem 2.22 is the key point for the foundational program: the ℓ_Q were assumed only finitely additive, but the theorem produces a σ-additive global measure whose marginals are exactly the ℓ_Q. In particular, each $\ell_Q = (\mathrm{eval}_Q)_\# P$ is σ-additive (as a pushforward of a σ-additive measure). Countable additivity is not an input — it is an output.

Corollary 2.24 (Architectural extension theorem)**.** *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system in which each outcome space O_Q is compact Hausdorff and each refinement map $\pi_{Q_1}^{Q_2}$ is continuous and surjective. Then every finitely compatible family $\{\ell_Q\}$ of finitely additive normalized valuations admits a unique σ-additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ with $(\mathrm{eval}_Q)_\# P = \ell_Q$ for every Q.*

Proof. By Proposition 2.9, compact outcome spaces and surjective refinement maps imply SPUT. Theorem 2.22 then applies directly. □

Remark 2.25 (Closing Stopping Point 2). Corollary 2.24 is the fully structural version of Stopping Point 2. Its hypotheses are entirely architectural: they concern the topology of the outcome spaces and the surjectivity of the refinement maps, not the valuations $\{\ell_Q\}$. Any observer whose query system has compact outcome spaces and surjective refinement maps is automatically in the setting where σ-additivity is derived from the architecture, not assumed. The observer need not check whether their laws are tight, or σ-additive, or anything beyond finite compatibility.

This is the strongest form of the result: *the structure of the observational interface alone forces countable additivity of the resulting global measure.*

Remark 2.26 (Hypothesis audit). Every hypothesis in Theorem 2.22 is used, and each at exactly one point:

Hypothesis	Where used	Can it be weakened?
$\mathcal{Q}$ countable	Concentration (§2.6)	Countable determining subsystem suffices
Sequentially upper-directed	Borel extension (§2.5)	Needed for closed-cylinder intersections
Hausdorff O_Q	Ω closed; graphs Borel	Essential
Continuous π	Ω closed; cylinders closed	Essential
Surjective bonding on K_Q	$K_\Omega \neq \emptyset$	Cannot weaken to containment
Compact K_Q	Tychonoff; FIP; Alexandrov	Essential
$\ell_Q(O_Q \setminus K_Q) < \varepsilon$	Mass estimates	Essential
Finite compatibility	μ_0 well-defined	Essential

2.8 Discussion

What the theorem achieves. Theorem 2.22 addresses Stopping Point 2 of the observational program: it shows that σ-additivity need not be assumed at each query level. Instead, it is a global consequence of two structural conditions: finite compatibility (the marginals cohere under refinement) and SPUT (the mass concentrates projectively on compatible compact cores). These conditions are properties of the *system* of observable laws, not of individual marginals.

The result is a strict weakening of the input to Paper 1: Paper 1 assumes σ-additive marginals and realizability, while this paper needs only finitely additive marginals and SPUT. The conclusion is the same: a σ-additive global measure on $(\Omega, \sigma(\mathcal{Q}))$.

What SPUT is buying. SPUT is a condition about how observable mass concentrates as one moves between query levels. It says, in operational terms: the things the observer considers *likely* at fine resolutions assemble coherently into things that are likely at coarser resolutions, and this assembly is non-degenerate (surjectivity). This is a natural condition on a query system arising from a physical process: if the system concentrates mass on a compact region at every observable resolution, that region is observable at all resolutions jointly.

What it costs. SPUT requires topological structure: Hausdorff spaces and continuous refinement maps. This is the price for starting from finitely additive marginals. Paper 1 operates without this topology, at the cost of requiring σ-additivity as an input. The two theorems occupy adjacent positions in the space of hypotheses, with this paper trading topology for weaker input measures.

The deeper question. Stopping Point 2 in its strongest form asks whether σ-additivity can be forced from *purely combinatorial* conditions on the query system, without topological assumptions. Theorem 2.22 does not achieve this: it replaces the assumption of σ-additivity with the assumption of Hausdorff topology plus SPUT. The question is whether SPUT itself can be derived from something more primitive.

The architectural theorem and the deeper question. Corollary 2.24 closes the first form of Stopping Point 2: in the compact outcome space setting with surjective refinement maps, σ-additivity is a theorem of architecture. The observer contributes only finite compatibility; the system does the rest.

This raises the next question: how restrictive is the compactness assumption? For sensor systems with bounded output spaces (voltages, pressures, counts in a finite range), compactness is natural. For infinite-range observables — real-valued sensor readings, delay vectors in $\mathbb{R}^d$, unbounded counts — it cannot be assumed. The question is what structural property of the outcome spaces substitutes for compactness.

The standard Borel route: Musiał's theorem. The existing measure-theoretic literature provides a second architectural resolution of Stopping Point 2, complementing Corollary 2.24 and covering the infinite-range case.

The key concept is *perfectness* in the sense of Gnedenko and Kolmogorov [3]: a probability measure μ on $(X, \mathcal{B})$ is *perfect* if for every measurable $f : X \to \mathbb{R}$, the image measure $f_*\mu$ is inner regular with respect to compact sets on $\mathbb{R}$. Perfectness is a measure-theoretic regularity condition that does not require topology on X itself.

The connection to the infinite-range case is given by two classical facts:

(i) *Standard Borel spaces carry only perfect measures.* If O_Q is a standard Borel space — a measurable space isomorphic to a Borel subset of a Polish space — then every Borel probability measure on O_Q is perfect [3, 6]. This follows from the Borel isomorphism theorem (any uncountable standard Borel space is measurably isomorphic to $\mathbb{R}$) together with inner regularity of finite Borel measures on $\mathbb{R}$.

(ii) *Projective systems of perfect measures have σ-additive limits with no leakage.* Musiał [8] proved that a projective system of perfect probability measures over any directed index set admits a unique σ-additive projective limit P satisfying $P(\Omega) = 1$. No topological conditions on the spaces or maps are required beyond measurability.

Combining these facts gives the following corollary, which is a consequence of [8] applied to the query system setting.

Corollary 2.27 (Standard Borel architectural extension)**.** *Let $(\mathcal{Q}, \preceq)$ be a countable sequentially upper-directed query system whose outcome spaces O_Q are standard Borel spaces and whose refinement maps are measurable. Then every finitely compatible family $\{\ell_Q\}$ of Borel probability measures on the query system extends uniquely to a σ-additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ with $P(\Omega) = 1$.*

Proof. Each ℓ_Q is a Borel probability measure on a standard Borel space, hence perfect by fact (i). Musiał's theorem (fact (ii)) gives a σ-additive projective limit on $\prod_Q O_Q$ with full mass on the coherent locus. The concentration argument of Section 2.6 applies: the graph of each measurable refinement map $\pi_{Q_1}^{Q_2} : O_{Q_2} \to O_{Q_1}$ is measurable in $O_{Q_1} \times O_{Q_2}$ when both spaces are standard Borel (the diagonal of a standard Borel space is measurable, since such spaces are T_2 and second countable after upgrading to their Polish topology [11]). Each defect set D_{Q_1,Q_2} is therefore measurable and null, and the countable union over all comparable pairs is null. □

What the two architectural corollaries together achieve. Corollaries 2.24 and 2.27 together resolve Stopping Point 2 in the following two architecturally natural settings:

Setting	Key condition	Corollary
Compact outcome spaces	Surjective refinement maps	2.24
Standard Borel outcome spaces	Measurable refinement maps	2.27

In both cases the conclusion is identical: σ-additivity is not an assumption on the observer's laws but a consequence of the query system's architecture. The observer need only supply finite compatibility; the spaces do the rest.

The standard Borel setting covers the main computational case of the program: delay query systems with outcome spaces $O_{Q_{d,\tau}} = \mathbb{R}^d$ are standard Borel, so any compatible family of stationary process marginals extends to a σ-additive global measure without any tightness assumption.

The leakage perspective. The Yosida–Hewitt theorem [5] provides the algebraic underpinning for both results. Every finitely additive probability measure ℓ on a measurable algebra decomposes uniquely as $\ell = \ell_c + \ell_p$, where ℓ_c is σ-additive and ℓ_p is *purely finitely additive* — the component that assigns positive mass to decreasing sequences of sets intersecting to empty. The *leakage* of a compatible family $\{\ell_Q\}$ is

$$\lambda(\{\ell_Q\}) := 1 - P(\Omega),$$

the total mass of the purely finitely additive parts that fails to concentrate on the coherent realization space Ω.

Stopping Point 2 asks when the query system architecture forces $\lambda = 0$ for every compatible family. The compact and standard Borel cases are two settings where this holds. The open question — the remaining form of Stopping Point 2 — is whether there are weaker architectural conditions on $(\mathcal{Q}, \preceq, \{\pi_{Q_1}^{Q_2}\})$ that also force $\lambda = 0$, applicable to spaces that are neither compact nor standard Borel.

Open problems.

1. *The topology-free route.* Both architectural corollaries use some structure on the outcome spaces: topology (compact Hausdorff) in Corollary 2.24, or measurable-space isomorphism to a Polish subspace (standard Borel) in Corollary 2.27. The fully general question is whether $\lambda = 0$ can be forced from purely order-theoretic conditions on $(\mathcal{Q}, \preceq)$, with no structural assumptions on the O_Q. The Mallory–Sion theorem [7] suggests the obstruction is measurability of the coherent locus Ω in the ambient product, and that this measurability is not guaranteed without additional structure.

2. *The uncountable case.* Concentration (Theorem 2.21) uses countability of $\mathcal{Q}$ to take a countable union of null defect sets. For uncountable $\mathcal{Q}$, the correct generalization is a condition on determining subsystems: if $\mathcal{Q}$ admits a countable determining subsystem $\mathcal{Q}_0$ (coherence on $\mathcal{Q}_0$ forces coherence on all of $\mathcal{Q}$), the argument applies. Characterizing when such subsystems exist is the open problem.

3. *Lean formalization.* The Lean 4/Mathlib formalization of this paper is nearly complete. The theorem `compactInverseLimit_nonempty` (Proposition 2.13(ii)) is fully proved via the finite intersection property on $\prod_Q K_Q$. The main theorem (`prokhorov_extension`) reduces to the algebraic extension theorem (`observational_extension` in `QuerySystem.lean`), which packages the Carathéodory construction, σ-subadditivity, marginal recovery, and uniqueness. One design point: the Lean statement of `prokhorov_extension` carries an explicit `EvalSurjective` hypothesis (surjectivity of eval_Q onto the full O_Q) not present in Theorem 2.22. This is because the algebraic route through `observational_extension` requires it directly, whereas the classical proof derives it from tightness via the inverse limit; a future refactor should add surjectivity of the bonding maps $\pi^{Q_2}_{Q_1}$ as a structural field and derive `EvalSurjective` from it. The one remaining `sorry` is `prokhorov_extension_polish` (Corollary 2.27): the notion of a perfect measure and Musiał's theorem are absent from Mathlib 4; formalizing this corollary would require adding both. The infrastructure for standard Borel spaces (the `StandardBorelSpace` typeclass) and inner regularity (the `Measure.InnerRegular` predicate) is available in Mathlib, providing the foundation for future formalization.

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From Extension to Determination: The Second Transition

Chapter 2 established that compatible observable laws determine a unique σ-additive probability measure P on $(\Omega, \sigma(\mathcal{Q}))$. The measure exists. It is canonical. Its marginals recover the original valuations $\{\ell_Q\}$.

But the construction was existential. We know P exists and is unique; we do not yet have a structural account of what P *is* — of how it is built from the query system, what properties it inherits from the $\{\ell_Q\}$, and when two different query systems determine the same P.

These are the questions of Chapter 3. Its central result is an *observational determination theorem*: two σ-additive probability measures on $(\Omega, \sigma(\mathcal{Q}))$ that agree on all query marginals are identical. Probability on the realization space is entirely determined by observable experiments — not partially, not generically, but exactly and completely.

This is the program's first structural theorem, and it has a philosophical weight beyond its technical content. It says that there is no hidden probability: no aspect of P that the observable interface fails to capture. The latent space Ω carries exactly the structure forced by the queries, and no more.

Chapter 3 also establishes two independent routes to the global measure, corresponding to two different starting hypotheses about the marginals: the *realizability route* (marginals already σ-additive, realizability condition on Ω) and the *Prokhorov route* (marginals only finitely additive, SPUT supplies σ-additivity). The relationship between these routes — and the conditions under which they coincide — clarifies the logical structure of what SP2 required.

The passage from Chapter 2 to Chapter 3 is the passage from *existence* to *structure*: from knowing that a probability exists to understanding how it is determined, when it is unique, and what observing it fully means.

Chapter 3

Observational Foundations of Probability

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Abstract

Many probabilistic theories begin by postulating an underlying probability space and then studying the observables derived from it. This work reverses that perspective. Starting from observable experiments organized as a refinement system of queries, we establish conditions under which compatible observable laws admit a canonical probability space representation.

We model observations as measurable queries with associated outcome spaces organized by a refinement preorder. From this system of queries we construct a canonical realization space as the projective limit of the observable outcome spaces. The observable σ-field is then generated by query-cylinder events.

Within this framework we establish two foundational results. First, probability measures on the observable σ-field are uniquely determined by the distributions of a generating family of queries, yielding an observational analogue of the classical π–λ uniqueness theorem. Second, compatible families of observable laws determine a canonical finitely additive observational content on cylinder events, and this content admits a probabilistic completion on the realization space by two independent routes.

The *realizability route* (Section 6) takes countably additive marginals as input and uses a realizability condition on Ω to produce the global measure via a direct Carathéodory argument, with no topology required on the outcome spaces. The *Prokhorov route* (Section 7) starts from only finitely additive marginals and derives countable additivity from a projective mass-concentration condition (surjective projective uniform tightness), at the cost of topological structure on the outcome spaces. Both routes yield a canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$ whose evaluation marginals recover the observable laws; under shared hypotheses they produce the same measure.

The resulting framework connects classical measure-theoretic probability, statistical experiment theory, and modern observable approaches to complex systems, and provides a foundation for extensions involving predictive structure, dynamical observables, and approximate compatibility.

3.1 Introduction and motivation

Probability theory traditionally begins with an underlying probability space and studies the observables derived from it. The present work reverses this perspective by starting from observable experiments themselves and asking when compatible observable laws admit a canonical probability space representation.

In many contemporary scientific settings the underlying event algebra is not directly accessible. Instead, systems are accessed through a family of admissible measurements, projections, or coarse-grainings. These observables determine the empirical distinctions that can be made about the system, while the latent event space remains implicit or even unknown. This raises a structural question: when can probabilistic structure be reconstructed from observable experiments alone?

Problem. When do observable experiments determine an underlying probability space?

Answer. We show that compatibility of observable laws is precisely the condition required for such a representation, yielding a canonical probability space whose evaluation marginals recover the observable experiments. The paper establishes this by two complementary routes. The realizability route requires: (i) measurable structure on each query outcome space; (ii) compatible countably additive probability laws; (iii) directedness conditions; and (iv) realizability of the projective limit. The Prokhorov route replaces hypothesis (ii) by finitely additive laws satisfying surjective projective uniform tightness, and derives σ-additivity from topological compactness of the outcome spaces. These two routes resolve the extension problem from opposite directions: one assumes countable coherence, the other derives it.

Program perspective. This work is part of a broader program developing dynamical structure from observable experiments. The present paper establishes the first stage: the representation theorem.

Classically, Kolmogorov's extension theorem constructs a probability measure from consistent finite-dimensional marginals. The present construction plays an analogous role for observable experiments: the query system determines a projective family of observable laws whose compatibility yields the canonical experiment $(\Omega, \mathcal{F}, P)$.

The structure of the framework is summarized by the following three-layer architecture.

$$
\begin{array}{cc}
\textbf{observational structure} & \textbf{probabilistic structure} \\
\text{queries } (\mathcal{Q},\preceq) & \\
\downarrow & \\
\text{projective system of outcomes } \{O_Q,\pi^{Q_2}_{Q_1}\} & \text{compatible observable laws } \{\nu_Q\} \\
\downarrow & \downarrow \\
\text{realization space } \Omega=\varprojlim O_Q & \textbf{finite observational content} \\
\downarrow & \downarrow \\
\text{observable } \sigma\text{-field } \sigma(\mathcal{Q}) & \text{probabilistic completion } P
\end{array}
$$

Main theorem. The principal result of this work establishes that compatible families of observable experiments admit a canonical probabilistic representation.

Theorem 3.1 (Observational representation theorem)**.** *Let* $(\mathcal{Q},\preceq)$ *be a query system equipped with a compatible family of observable laws (under the structural hypotheses of Sections 3 and 6: measurable outcome spaces, countably additive laws, directedness, and realizability). Then there exists a canonical probability space*

$$(\Omega,\mathcal{F},P)$$

such that for every query $Q\in\mathcal{Q}$ *the evaluation map*

$$Q:\Omega\to O_Q$$

recovers the corresponding observable law.

Relation to the predictive sequel. The canonical experiment constructed here provides the probabilistic representation of observable experiments. A subsequent work studies how predictive state spaces and operator dynamics arise naturally from this observational representation.

Scope. The present paper does not derive local measurable structure or local countable additivity from observation alone. Rather, it shows that once queries are equipped with measurable outcome spaces and compatible countably additive observable laws, the remaining global probabilistic structure is canonically forced under explicit structural assumptions on the query system. The derivation of local probabilistic structure from more primitive observable distinctions is a direction for future work.

Structure of the paper. Section 2 recalls the classical uniqueness and extension principles that motivate the construction. Section 3 introduces query systems, their realization spaces, and the observable σ-field, together with the three structural conditions used in the main results. Section 4 proves the observational determination theorem. Section 5 constructs the finite observational content from compatible observable laws. Section 6 proves the observational extension theorem by a direct Carathéodory argument, establishing the canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$ without topological assumptions on the outcome spaces. Section 7 establishes the Prokhorov extension theorem: a parallel route in which σ-additivity is derived rather than assumed, under surjective projective uniform tightness of the marginals. The final sections interpret the framework in relation to statistical experiment theory and outline directions for predictive and dynamical extensions.

3.2 Classical uniqueness and extension

A central structural feature of classical probability theory is that a probability measure is determined by its values on a generating class of events. One of the most widely used tools expressing this fact is the π–λ theorem [1, 6]. Let $(\Omega, \mathcal{F})$ be a measurable space, and let $\mathcal{C} \subset \mathcal{F}$ be a π-system, that is, a collection of sets closed under finite intersections. If $\mathcal{C}$ generates $\mathcal{F}$, then any two probability measures P and P' satisfying

$$P(A) = P'(A), \qquad A \in \mathcal{C},$$

necessarily agree on the full σ-algebra:

$$P = P' \text{ on } \mathcal{F}.$$

In this sense, the global probabilistic structure is determined by its values on a suitable generating class.

A particularly important application of this principle is Kolmogorov's extension theorem [7, 1]. In the classical construction of stochastic processes, one begins by specifying a single-time state space $(S, \mathcal{S})$ together with an index set T, typically representing time. The trajectory space is then taken to be the product

$$\Omega = S^T,$$

equipped with the product σ-algebra

$$\mathcal{F} = \mathcal{S}^{\otimes T}.$$

The measurable structure is generated by the coordinate projections

$$\pi_t : \Omega \to S, \qquad \pi_t(\omega) = \omega(t).$$

More generally, for finite subsets $\{t_1, \ldots, t_n\} \subset T$, one considers the joint projections

$$\pi_{t_1,\ldots,t_n}(\omega) = \big(\omega(t_1), \ldots, \omega(t_n)\big),$$

which generate the cylinder σ-field.

A stochastic process is then specified by a compatible family of finite-dimensional distributions

$$\mu_{t_1,\ldots,t_n} \in \mathcal{P}(S^n),$$

satisfying the marginal consistency condition

$$(\pi_{t_1,\ldots,t_n})_{\#}\mu_{t_1,\ldots,t_m} = \mu_{t_1,\ldots,t_n}, \qquad n < m.$$

Kolmogorov's theorem asserts that such a compatible family determines a unique probability measure

$$P \in \mathcal{P}(\Omega, \mathcal{F})$$

whose finite-dimensional marginals coincide with the prescribed laws.

This construction makes a specific structural assumption about the organization of observational information. The starting point is a global trajectory space together with a canonical system of coordinate projections. Measurable events are generated by restrictions of these coordinates, and probabilistic structure is built relative to this coordinate representation.

However, in many contemporary scientific and statistical settings, this assumption is neither natural nor directly available. Systems are typically accessed through families of admissible measurements, projections, or coarse-grainings, without prior knowledge of a global coordinate structure. The observable interface is therefore given by measurable maps rather than by coordinates of an assumed latent space.

This observation suggests a reformulation in which admissible observations are taken as primitive. Instead of postulating a measurable event space in advance, we derive the observable σ-field from a family of measurable maps and formulate probabilistic structure in terms of compatibility of their induced laws.

3.3 Observational generators

Replacing coordinates by a more general notion of observable structure requires specifying the observational interface directly. Rather than beginning with a fixed measurable state space, we start from a family of admissible queries.

Definition 3.2 (Query)**.** A *query* Q consists of a measurable outcome space

$$(O_Q, \mathcal{B}(O_Q)).$$

Remark 3.3 (Primitive structure). In the present paper the measurable structure $\mathcal{B}(O_Q)$ on each outcome space is taken as a primitive input. It represents the class of events distinguishable at resolution Q. A more foundational treatment would derive this σ-algebra from a designated class $\mathcal{E}_Q$ of admissible observable distinctions via $\mathcal{B}(O_Q) = \sigma(\mathcal{E}_Q)$; we regard this as a direction for future work.

A family of queries $\mathcal{Q}$ represents the admissible observational interface through which the system is accessed.

Definition 3.4 (Query system). A *query system* consists of:

1. a preorder $(\mathcal{Q}, \preceq)$;
2. for each $Q \in \mathcal{Q}$, a measurable outcome space
$$(O_Q, \mathcal{B}(O_Q));$$
3. for each relation $Q_1 \preceq Q_2$, a measurable refinement map
$$\pi_{Q_1}^{Q_2} : O_{Q_2} \to O_{Q_1},$$
satisfying the coherence relations
$$\pi_Q^Q = \mathrm{id}_{O_Q}, \qquad \pi_{Q_1}^{Q_2} \circ \pi_{Q_2}^{Q_3} = \pi_{Q_1}^{Q_3} \quad (Q_1 \preceq Q_2 \preceq Q_3).$$

In this way a query system $(\mathcal{Q}, \preceq)$ may be interpreted as an information order in the sense of Blackwell and Le Cam [2, 9]: queries higher in the order are at least as informative as those below them, since the outcomes of coarser queries are obtained from finer ones by measurable post-processing. Thus the refinement maps are coarsening maps: whenever $Q_1 \preceq Q_2$, the more informative query Q_2 determines the outcome of the less informative query Q_1.

We will use two structural hypotheses, depending on the goal.

Definition 3.5 (Common coarsenings / lower-directedness). We say $(\mathcal{Q}, \preceq)$ admits *common coarsenings* if for any $Q_1, Q_2 \in \mathcal{Q}$ there exists $Q_3 \in \mathcal{Q}$ such that
$$Q_3 \preceq Q_1 \qquad \text{and} \qquad Q_3 \preceq Q_2.$$

Intuitively, Q_3 is a single (less informative) query whose outcomes can be pushed forward to recover the outcomes of both Q_1 and Q_2. This is the assumption that any two observational resolutions admit a joint coarser representation.

Definition 3.6 (Common refinements / upper-directedness). We say $(\mathcal{Q}, \preceq)$ admits *common refinements* if for any $Q_1, Q_2 \in \mathcal{Q}$ there exists $Q_3 \in \mathcal{Q}$ such that
$$Q_1 \preceq Q_3 \qquad \text{and} \qquad Q_2 \preceq Q_3.$$

This is the assumption that any two observational resolutions can be jointly refined by a third — an assumption about the richness of the observational interface.

Definition 3.7 (Sequential common refinements)**.** We say $(\mathcal{Q}, \preceq)$ admits *sequential common refinements* if every countable family $Q_1, Q_2, \ldots \in \mathcal{Q}$ admits a common refinement: there exists $Q_* \in \mathcal{Q}$ such that

$$Q_n \preceq Q_* \qquad \text{for all } n \geq 1.$$

This is the assumption that the observational interface is closed under countable joint refinement: however many query resolutions are specified, a single finer resolution exists that subsumes all of them. This is a substantive condition on the interface structure, not merely a technical regularity hypothesis.

Sequential common refinements imply common refinements (take the pair Q_1, Q_2 as the sequence). The sequential condition is strictly stronger and is the key analytic hypothesis enabling σ-subadditivity of the cylinder premeasure.

Definition 3.8 (Realizability)**.** The query system is *realizable* if every local outcome extends to a global realization: for every $Q \in \mathcal{Q}$ and every $o \in O_Q$, there exists $\omega \in \Omega$ such that $\mathrm{eval}_Q(\omega) = o$.

Equivalently, the evaluation maps $\mathrm{eval}_Q : \Omega \to O_Q$ are all surjective.

Remark 3.9 (Status of realizability). Realizability is a substantive assumption. It asserts that the projective limit Ω is non-degenerate in a strong pointwise sense: every locally consistent outcome extends to a globally coherent realization. This is used in an essential way in the premeasure well-definedness argument and in the σ-subadditivity step of the extension theorem — it is not a side condition. Without it, both steps fail.

A natural measure-theoretic weakening — likely sufficient for the proof — is *almost-sure realizability*: for each Q and ν_Q-almost every $o \in O_Q$, there exists $\omega \in \Omega$ with $\mathrm{eval}_Q(\omega) = o$. We retain the stronger pointwise form here for clarity. A sufficient condition for full realizability is given in Corollary 6.1: when each O_Q is Polish, refinement maps are continuous, and the index set is countable, the projective limit is Polish and realizability holds automatically.

Both structural hypotheses are used in the main results. Common coarsenings are used for the uniqueness theorem (the cylinder family forms a π-system). Common refinements are used for the extension theorem (finite cylinder families can be compressed to a single query level to define a well-posed premeasure). A sequential strengthening of common refinements is the key analytic condition that allows the premeasure to be extended to a full probability measure.

The space of realizations is defined as the set of coherent answers to all queries:

$$\Omega = \left\{ x \in \prod_{Q \in \mathcal{Q}} O_Q \;\middle|\; x_{Q_1} = \pi^{Q_2}_{Q_1}(x_{Q_2}) \text{ whenever } Q_1 \preceq Q_2 \right\}.$$

Equivalently,

$$\Omega = \varprojlim_{Q \in \mathcal{Q}} O_Q,$$

the projective limit of the system of outcome spaces with bonding maps $\pi^{Q_2}_{Q_1}$ [11].

Remark 3.10 (Nondegeneracy of Ω). The realization space Ω is a derived object: it is determined entirely by the query system and carries no structure beyond what the refinement maps force. It is not assumed as a latent state space. However, Ω can be empty or degenerate without additional hypotheses on the query system — realizability (Definition 3.8) is precisely the condition asserting non-degeneracy.

The realization space Ω serves as a canonical space of coherent answers to all admissible queries. Probability measures on Ω represent probabilistic models compatible with the observable interface.

Each query Q induces an evaluation map

$$\mathrm{eval}_Q : \Omega \to O_Q, \qquad \mathrm{eval}_Q(x) = x_Q.$$

Probability measures on Ω represent observational laws compatible with all admissible queries; the pushforwards $(\mathrm{eval}_Q)_\# P$ recover the observable marginal distributions.

The distribution of a query under a probability measure $P \in \mathcal{P}(\Omega, \sigma(\mathcal{Q}))$ is given by the pushforward measure

$$Q_\# P = (\mathrm{eval}_Q)_\# P \in \mathcal{P}(O_Q).$$

Lemma 3.11 (Projection compatibility)**.** *For every relation $Q_1 \preceq Q_2$ in $\mathcal{Q}$ the evaluation maps satisfy*

$$\mathrm{eval}_{Q_1} = \pi^{Q_2}_{Q_1} \circ \mathrm{eval}_{Q_2}.$$

Proof. For $x \in \Omega$ we have

$$(\pi^{Q_2}_{Q_1} \circ \mathrm{eval}_{Q_2})(x) = \pi^{Q_2}_{Q_1}(x_{Q_2}).$$

By the defining compatibility condition of Ω,

$$x_{Q_1} = \pi^{Q_2}_{Q_1}(x_{Q_2}),$$

so the right-hand side equals $\mathrm{eval}_{Q_1}(x)$. $\square$

The observable σ-field is defined as the smallest σ-algebra making all evaluation maps measurable. Equivalently, it is generated by the cylinder sets

$$\mathrm{Cyl}(Q, A) = \{x \in \Omega : \mathrm{eval}_Q(x) \in A\}, \qquad A \in \mathcal{B}(O_Q).$$

We denote this σ-field by

$$\sigma(\mathcal{Q}).$$

Lemma 3.12 (Measurability of evaluation maps)**.** *For every query $Q \in \mathcal{Q}$, the evaluation map*

$$\mathrm{eval}_Q : \Omega \to O_Q$$

is measurable with respect to $(\Omega, \sigma(\mathcal{Q}))$.

Proof. By definition $\sigma(\mathcal{Q})$ is the smallest σ-algebra making all evaluation maps measurable. □

The common-coarsening hypothesis introduced above is a meet-like property of the information order on queries. Intuitively, it ensures that finite collections of observable constraints can be represented at a single query level.

Lemma 3.13 (Cylinder π-system)**.** *Assume $(\mathcal{Q}, \preceq)$ admits common coarsenings in the sense of Definition 3.5.*

Then the family of cylinder sets

$$\mathcal{C}_{\mathcal{Q}} = \{\mathrm{Cyl}(Q, A) : Q \in \mathcal{Q},\ A \in \mathcal{B}(O_Q)\}$$

forms a π-system.

Proof. Let $\mathrm{Cyl}(Q_1, A_1)$ and $\mathrm{Cyl}(Q_2, A_2)$ be two cylinders. The common-coarsening hypothesis provides a query Q_3 such that

$$Q_3 \preceq Q_1 \qquad \text{and} \qquad Q_3 \preceq Q_2.$$

Using the refinement maps $\pi^{Q_1}_{Q_3}$ and $\pi^{Q_2}_{Q_3}$, a direct calculation shows

$$\mathrm{Cyl}(Q_1, A_1) \cap \mathrm{Cyl}(Q_2, A_2) = \mathrm{Cyl}\Big(Q_3, (\pi^{Q_1}_{Q_3})^{-1}(A_1) \cap (\pi^{Q_2}_{Q_3})^{-1}(A_2)\Big).$$

Since measurable sets are stable under preimages and intersections, the right-hand side is again a cylinder. □

Remark 3.14 (Relation to classical coordinate constructions)*.* In the classical Kolmogorov construction of stochastic processes, the indexing family of coordinate projections is ordered by inclusion of finite coordinate sets and is therefore upper-directed. The observational framework uses the opposite side of the information order: common coarsenings permit finite collections of observable constraints to be represented at a single query level.

3.4 Observational determination

The following result is the observational analogue of the classical π–λ uniqueness theorem.

Theorem 3.15. *Let $(\mathcal{Q}, \preceq)$ admit common coarsenings (Definition 3.5) and let P and P' be probability measures on the observable space $(\Omega, \sigma(\mathcal{Q}))$ generated by $\mathcal{Q}$.*

If

$$Q_\# P = Q_\# P'$$

for all $Q \in \mathcal{Q}$, then $P = P'$.

Proof. Equality of pushforward laws implies that P and P' agree on all cylinder events. By Lemma 3.13 the cylinder family forms a π-system generating $\sigma(\mathcal{Q})$. The classical π–λ theorem therefore implies that the two measures coincide on the entire observable σ-field. □

Remark 3.16. The determination theorem is purely measure-theoretic. It depends only on the observable σ-field generated by queries and on the lower-directedness hypothesis ensuring that cylinder events form a π-system. No compactness or topological assumptions enter at this stage.

3.5 Finite observational content

Definition 3.17 (Cylinder set semiring)**.** Let

$$\mathcal{C}_{\mathcal{Q}} = \{\mathrm{Cyl}(Q, A) : Q \in \mathcal{Q},\ A \in \mathcal{B}(O_Q)\}$$

denote the family of cylinder sets.

Lemma 3.18 (Cylinder set semiring structure)**.** *Assume* $(\mathcal{Q}, \preceq)$ *admits common coarsenings (Definition 3.5). Then* $\mathcal{C}_{\mathcal{Q}}$ *is a set semiring: it is closed under finite intersections (already the* π*-system property of Lemma 3.13) and set differences decompose as finite disjoint unions of cylinders.*

Proof sketch. Intersection: Lemma 3.13. Difference: given $\mathrm{Cyl}(Q_1, A) \setminus \mathrm{Cyl}(Q_2, B)$, use a common coarsening $Q_3 \preceq Q_1, Q_2$ to write both as cylinders at Q_3; the difference is then a single cylinder at Q_3 and is itself in $\mathcal{C}_{\mathcal{Q}}$. □

Lemma 3.19 (Finite cylinder compression)**.** *Assume* $(\mathcal{Q}, \preceq)$ *admits common refinements (Definition 3.6).*

Then for any finite family of queries $Q_1, \ldots, Q_n$ *and measurable sets* $A_i \in \mathcal{B}(O_{Q_i})$*, there exists a common refinement* Q_* *such that*

$$\bigcap_{i=1}^{n} \mathrm{Cyl}(Q_i, A_i) = \mathrm{Cyl}\left(Q_*,\ \bigcap_{i=1}^{n} (\pi_{Q_i}^{Q_*})^{-1}(A_i)\right).$$

Proof. Upper-directedness gives a common refinement Q_* with $Q_i \preceq Q_*$ for all i. Then $\pi_{Q_i}^{Q_*} : O_{Q_*} \to O_{Q_i}$ is well-typed and $\mathrm{Cyl}(Q_i, A_i) = \mathrm{Cyl}(Q_*, (\pi_{Q_i}^{Q_*})^{-1}(A_i))$ by the refinement identity. The intersection follows. □

Remark 3.20. Lemma 3.19 uses common *refinements* rather than common coarsenings. The direction matters: the projection $\pi_{Q_i}^{Q_*} : O_{Q_*} \to O_{Q_i}$ goes from the finer space to the coarser one, so it is the preimage that lives in O_{Q_*} and the constraint set is well-typed. An upper bound in the information order is what is needed here.

We now turn to the question of probabilistic realization. Suppose we are given, for each query $Q \in \mathcal{Q}$, a probability law

$$\nu_Q \in \mathcal{P}(O_Q).$$

Definition 3.21 (Observational compatibility)**.** A family $\{\nu_Q\}_{Q\in\mathcal{Q}}$ with $\nu_Q \in \mathcal{P}(O_Q)$ is *compatible* if for every relation $Q_1 \preceq Q_2$ in the query preorder,

$$(\pi^{Q_2}_{Q_1})_{\#}\nu_{Q_2} = \nu_{Q_1}.$$

In other words, whenever one query is obtained from another by coarsening, the corresponding probability laws agree under the induced marginalization.

Theorem 3.22 (Finite observational content)**.** *Suppose:*

1. *The refinement preorder on $\mathcal{Q}$ admits common coarsenings (Definition 3.5) and common refinements (Definition 3.6).*
2. *The query system is realizable (Definition 3.8).*
3. *The family $\{\nu_Q\}_{Q\in\mathcal{Q}}$ is observationally compatible in the sense of Definition 3.21.*

Then the family $\{\nu_Q\}$ determines a well-defined finitely additive content μ_0 on $\mathcal{C}_{\mathcal{Q}}$:

$$\mu_0(\mathrm{Cyl}(Q, A)) := \nu_Q(A), \qquad A \in \mathcal{B}(O_Q).$$

Proof sketch. Well-definedness. Suppose $\mathrm{Cyl}(Q, A) = \mathrm{Cyl}(Q', A')$. Use common refinements to find Q_* finer than both Q and Q'. The constraint sets at Q_* are the preimages of A and A' under $\pi^{Q_*}_{Q}$ and $\pi^{Q_*}_{Q'}$ respectively. Since $\mathrm{Cyl}(Q, A) = \mathrm{Cyl}(Q', A')$, surjectivity of eval_{Q_*} (realizability) implies the two preimage sets in O_{Q_*} coincide. Compatibility then gives $\nu_Q(A) = \nu_{Q_*}(\text{preimage}) = \nu_{Q'}(A')$.

Finite additivity. For disjoint cylinders with measurable union, the same compression to a common refinement level Q_* reduces additivity to that of ν_{Q_*}. □

Remark 3.23. The passage from finite content to a countably additive probability measure on $(\Omega, \sigma(\mathcal{Q}))$ requires one further condition: σ-subadditivity of μ_0. This is the analytic heart of the extension problem, and the next section shows how sequential common refinements supply it.

Hypothesis (3) requires each ν_Q to be a countably additive probability measure. This is assumed, not derived: the countable additivity present in the global extension theorem (§6) is propagated from this local assumption via the compression argument, not constructed from below. The question of whether countable additivity of the ν_Q can itself be derived from compatibility and interface structure alone is a direction for future work.

3.6 Observational extension theorem

The finite content μ_0 constructed in the previous section is finitely additive on the cylinder set semiring $\mathcal{C}_{\mathcal{Q}}$. The central result of this paper shows that sequential common refinements supply the missing σ-subadditivity, allowing Carathéodory's extension theorem [1, 3] to produce a full probability measure on $(\Omega, \sigma(\mathcal{Q}))$.

Theorem 3.24 (Observational extension theorem)**.** *Suppose:*

1. *$(\mathcal{Q}, \preceq)$ admits common coarsenings (Definition 3.5).*
2. *$(\mathcal{Q}, \preceq)$ admits sequential common refinements (Definition 3.7).*
3. *The query system is realizable (Definition 3.8).*
4. *The family $\{\nu_Q\}_{Q\in\mathcal{Q}}$ is observationally compatible (Definition 3.21) and each ν_Q is a probability measure.*

Then there exists a unique probability measure

$$P \in \mathcal{P}(\Omega, \sigma(\mathcal{Q}))$$

such that for every $Q \in \mathcal{Q}$,

$$(\mathrm{eval}_Q)_\# P = \nu_Q.$$

Proof. Step 1: Finite content. Theorem 3.22 gives a well-defined finitely additive content μ_0 on $\mathcal{C}_{\mathcal{Q}}$ with $\mu_0(\mathrm{Cyl}(Q, A)) = \nu_Q(A)$. By Lemma 3.18, $\mathcal{C}_{\mathcal{Q}}$ is a set semiring.

Step 2: σ-subadditivity. Let $\mathrm{Cyl}(Q, B) \subseteq \bigcup_{n\geq 1} \mathrm{Cyl}(Q_n, A_n)$ with all sets in $\mathcal{C}_{\mathcal{Q}}$. Apply sequential common refinements to the sequence $Q, Q_1, Q_2, \ldots$ to obtain a single query Q_* with $Q \preceq Q_*$ and $Q_n \preceq Q_*$ for all n.

Compression (Lemma 3.19) rewrites each cylinder at level Q_*:

$$\mathrm{Cyl}(Q, B) = \mathrm{Cyl}(Q_*, E), \qquad \mathrm{Cyl}(Q_n, A_n) = \mathrm{Cyl}(Q_*, E_n),$$

where $E = (\pi_Q^{Q_*})^{-1}(B)$ and $E_n = (\pi_{Q_n}^{Q_*})^{-1}(A_n)$ are measurable subsets of O_{Q_*}. Realizability and the set inclusion in Ω imply $E \subseteq \bigcup_n E_n$ in O_{Q_*}. Therefore

$$\mu_0(\mathrm{Cyl}(Q, B)) = \nu_{Q_*}(E) \leq \sum_{n=1}^{\infty} \nu_{Q_*}(E_n) = \sum_{n=1}^{\infty} \mu_0(\mathrm{Cyl}(Q_n, A_n)),$$

where the inequality is σ-subadditivity of the single measure ν_{Q_*}.

Step 3: Carathéodory extension. Since $\mathcal{C}_{\mathcal{Q}}$ is a set semiring generating $\sigma(\mathcal{Q})$ and μ_0 is a σ-subadditive additive content on it, Carathéodory's extension theorem [1, 3] applies and yields a measure P on $(\Omega, \sigma(\mathcal{Q}))$ extending μ_0.

Step 4: Marginal recovery. For any $A \in \mathcal{B}(O_Q)$,

$$P(\mathrm{eval}_Q^{-1}(A)) = P(\mathrm{Cyl}(Q, A)) = \mu_0(\mathrm{Cyl}(Q, A)) = \nu_Q(A),$$

so $(\mathrm{eval}_Q)_\# P = \nu_Q$.

Step 5: P is a probability measure. $P(\Omega) = P(\mathrm{Cyl}(Q, O_Q)) = \nu_Q(O_Q) = 1$ for any Q.

Uniqueness. Two probability measures with the same evaluation marginals agree on every cylinder event, hence on $\sigma(\mathcal{Q})$ by Theorem 3.15. □

Remark 3.25 (Assumptions versus derived structure). The theorem assumes: (i) measurable structure on each outcome space; (ii) countable additivity of each observable law ν_Q; (iii) lower-directedness, sequential upper-directedness, and realizability of the query system. It derives: a canonical σ-additive probability measure on $(\Omega, \sigma(\mathcal{Q}))$ whose evaluation marginals recover all observable laws.

Countable additivity of P is not constructed from first principles but propagated from the countable additivity of ν_{Q_*} via the compression argument. Assumptions (i) and (ii) are inputs from classical measure theory; a more foundational treatment would aim to derive them from the structure of observable distinctions. Assumptions (iii) are substantive conditions on the query system: they express joint coarsenability, closure under countable joint refinement, and non-degeneracy of the realization space.

Remark 3.26 (Structure of the proof). The proof uses the two directedness conditions for distinct roles. Common coarsenings give the semiring structure on $\mathcal{C}_\mathcal{Q}$ (needed for Carathéodory) and the π-system structure needed for uniqueness. Sequential common refinements compress any countable cylinder family to a single query level, reducing σ-subadditivity of μ_0 to σ-subadditivity of the single measure ν_{Q_*}, which is immediate. Realizability is used only to translate set inclusions in Ω to set inclusions in O_{Q_*}.

Remark 3.27 (Relation to Kolmogorov's extension theorem). Kolmogorov's theorem constructs a probability measure on S^T from consistent finite-dimensional distributions. The index set is a coordinate product and the consistency condition is the push-forward condition under coordinate projections.

The observational extension theorem plays the same role for query systems. The realization space $\Omega = \varprojlim O_Q$ replaces the product S^T, compatible marginals replace consistent finite-dimensional distributions, and sequential common refinements replace the countable-cofinality condition that reduces the Kolmogorov argument to a countable system.

The key conceptual difference is that the observational route imposes no topology on the outcome spaces. The entire construction is measure-theoretic: the measure P is built directly on the $\sigma(\mathcal{Q})$-measurable structure of Ω by Carathéodory's theorem, without any appeal to tightness or compactness.

Corollary 3.28 (Topological setting)**.** *In particular, Corollary 3.28 provides the main route to verifying realizability: under Polish outcome spaces and a countable cofinal chain, all four hypotheses of Theorem 3.24 are automatically satisfied.*

The hypotheses of Theorem 3.24 are satisfied whenever:

1. *each outcome space O_Q is Polish and the refinement maps are continuous (so Ω is a closed subspace of a countable product of Polish spaces, hence Polish [3] and realizable);*

2. *the query preorder contains a countable cofinal chain $Q_1 \preceq Q_2 \preceq \cdots$ (so every sequence of queries admits a common refinement within this chain).*

In this case the compatible observable laws need not be assumed Radon: the measure P produced by Theorem 3.24 is automatically a Radon probability measure on the Polish space Ω.

Remark 3.29 (Lean formalization). The proof of Theorem 3.24 given above is the proof that is fully formalized in Lean 4 [5] (`observational_extension` in `QuerySystem.lean`, 0 sorrys). The Lean hypotheses correspond exactly to the four conditions of the theorem: `LowerDirected` (common coarsenings), `SequentiallyUpperDirected` (sequential common refinements), `EvalSurjective` (realizability), and `CompatibleMarginals` with `IsProbabilityMeasure`. The Carathéodory step uses `AddContent.measure` from Mathlib [12].

3.7 Prokhorov extension for query systems

Theorem 3.24 proceeds from the *realizability* side: given σ-additive marginals $\{\nu_Q\}$ and a realizability condition on Ω, it produces the global measure P. This section establishes a route from the opposite direction, the *tightness* side: given only *finitely* additive marginals, a projective mass-concentration condition forces σ-additivity and delivers P directly. These two theorems resolve the same extension problem from opposite directions: one assumes countable coherence, the other derives it.

Projective uniform tightness

Because the marginals $\{\nu_Q\}$ live on different spaces O_Q, the classical Prokhorov notion of tightness cannot be applied level by level. The correct formulation is projective: compact approximants must be chosen compatibly across levels so that they assemble into a compact subset of the inverse limit Ω.

Definition 3.30 (Surjective projective uniform tightness)**.** A finitely compatible family $\{\nu_Q\}_{Q\in\mathcal{Q}}$ of finitely additive normalized measures on $(\mathcal{B}(O_Q))_{Q\in\mathcal{Q}}$ is *surjectively projectively uniformly tight* if for every $\varepsilon > 0$ there exists a system $\{K_Q\}_{Q\in\mathcal{Q}}$ satisfying:

1. $K_Q \subset O_Q$ is compact for each $Q \in \mathcal{Q}$,

2. $\pi^{Q_2}_{Q_1}(K_{Q_2}) = K_{Q_1}$ for all $Q_1 \preceq Q_2$ *(surjectivity of bonding maps on compact sets)*,

3. $\nu_Q(O_Q \setminus K_Q) < \varepsilon$ for every $Q \in \mathcal{Q}$.

Condition (2) ensures the compact sets form a consistent inverse system, so their intersection in Ω,

$$K_\Omega := \Omega \cap \prod_{Q \in \mathcal{Q}} K_Q = \varprojlim K_Q,$$

is nonempty and compact, and each projection $\mathrm{eval}_Q : K_\Omega \to K_Q$ is surjective (by the inverse-limit theorem for compact Hausdorff spaces with surjective bonding maps).

Remark 3.31 (Relation to individual tightness)*.* Definition 3.30 strengthens individual tightness (compact K_Q at each level independently) by requiring the compact approximants to be chosen compatibly under refinement with surjective bonding maps. This cross-level compatibility is precisely what is needed to lift a local nonemptiness argument into Ω.

The theorem

Theorem 3.32 (Prokhorov extension for query systems)**.** *Let* $(\mathcal{Q}, \preceq)$ *be a countable sequentially upper-directed query system in which each outcome space* O_Q *is Hausdorff with* $\mathcal{E}_Q = \mathcal{B}(O_Q)$*, and each refinement map* $\pi^{Q_2}_{Q_1} : O_{Q_2} \to O_{Q_1}$ *is continuous. Let* $\{\nu_Q\}_{Q \in \mathcal{Q}}$ *be a finitely compatible family of finitely additive normalized measures on* $\mathcal{B}(O_Q)$*.*

If $\{\nu_Q\}$ *is surjectively projectively uniformly tight (Definition 3.30), then there exists a unique* σ*-additive probability measure* P *on* $(\Omega, \sigma(\mathcal{Q}))$ *satisfying*

$$(\mathrm{eval}_Q)_\# P = \nu_Q \qquad \textit{for every } Q \in \mathcal{Q}.$$

Uniqueness is among all σ*-additive probability measures on* $(\Omega, \sigma(\mathcal{Q}))$ *with the prescribed marginals.*

Proof sketch. Step 1: Compact core. For each $\varepsilon > 0$ apply Definition 3.30 to obtain $\{K_Q\}$ with surjective bonding maps, compact $K_\Omega = \varprojlim K_Q$, and $\nu_Q(O_Q \setminus K_Q) < \varepsilon$ for all Q.

Step 2: Closed-cylinder continuity from above. Let $C_N = \mathrm{Cyl}(Q_N, F_N) \downarrow \emptyset$ with each $F_N \subseteq O_{Q_N}$ closed. Assuming $\mu_0(C_N) \geq \varepsilon$ for all N (seeking contradiction), the mass estimate $\nu_{Q_N}(F_N \cap K_{Q_N}) \geq \varepsilon/2 > 0$ combined with surjectivity of $\mathrm{eval}_{Q_N} : K_\Omega \to K_{Q_N}$ gives $C_N \cap K_\Omega \neq \emptyset$ for all N. Since F_N is closed and eval_{Q_N} is continuous, each $C_N \cap K_\Omega$ is closed in the compact space K_Ω. The finite intersection property yields $\bigcap_N (C_N \cap K_\Omega) \neq \emptyset$, contradicting $C_N \downarrow \emptyset$ in Ω.

Step 3: Extension to Borel cylinders. Each K_Q is compact Hausdorff, so $\nu_Q|_{K_Q}$ is regular by Alexandrov's theorem (no additional hypothesis required). Inner regularity approximates every Borel cylinder by closed cylinders to within $O(\varepsilon)$. Sequential upper-directedness provides common refinements for finite intersections of closed cylinders, preserving closedness. A standard approximation argument reduces the general Borel case to the closed-cylinder case already handled. Carathéodory's theorem then extends the σ-additive premeasure μ_0 uniquely to $\sigma(\mathcal{C}_{\mathcal{Q}}) = \sigma(\mathcal{Q})$ on the ambient product.

Step 4: Concentration on Ω (graph-support lemma). For each comparable pair $Q_1 \preceq Q_2$, the intersection $\mathrm{Cyl}(Q_1, A) \cap \mathrm{Cyl}(Q_2, B) = \mathrm{Cyl}(Q_2, (\pi^{Q_2}_{Q_1})^{-1}(A) \cap B)$ shows the joint law of $(\mathrm{eval}_{Q_1}, \mathrm{eval}_{Q_2})$ under P equals $(\mathrm{graph}_\pi)_\# \nu_{Q_2}$, which is supported on the graph $G_{Q_1,Q_2} = \{(a, b) : a = \pi^{Q_2}_{Q_1}(b)\}$. Hence the defect set $D_{Q_1,Q_2} := \{x : x_{Q_1} \neq \pi^{Q_2}_{Q_1}(x_{Q_2})\}$ has $P(D_{Q_1,Q_2}) = 0$. Since $\mathcal{Q}$ is countable there are countably many comparable pairs; a countable union of null sets gives $P(\Omega) = 1$. Restricting P to Ω yields the required measure. Uniqueness follows because cylinder sets generate $\sigma(\mathcal{Q})$ and determine any σ-additive probability measure with the given marginals. $\square$

Proposition 3.33 (Consistency between routes)**.** *If $\{\nu_Q\}$ is a family of σ-additive probability measures satisfying both the hypotheses of Theorem 3.24 and those of Theorem 3.32, then both constructions produce the same measure P.*

Proof. Both measures are σ-additive on $(\Omega, \sigma(\mathcal{Q}))$ with marginals ν_Q for every Q. Since cylinder sets generate $\sigma(\mathcal{Q})$ and form a π-system, uniqueness of σ-additive extensions implies equality. $\square$

Remark 3.34 (Comparison of the two routes)*.* The two extension theorems address complementary scenarios.

Route	Input	Key mechanism
Realizability (Thm. 3.24)	σ-additive ν_Q	realizability of Ω
Prokhorov (Thm. 3.32)	finitely additive ν_Q	projective tightness

The realizability route assumes countable coherence of the marginals and requires no topology on the outcome spaces. The Prokhorov route derives countable coherence from a topological mass-concentration condition: surjective projective compact cores prevent mass from escaping the inverse limit. Both routes conclude that a unique global measure P on $(\Omega, \sigma(\mathcal{Q}))$ exists with the prescribed marginals.

In practice, the realizability route applies when countable additivity of the marginals is known a priori; the Prokhorov route applies when only finite compatibility is available but surjective projective uniform tightness (Definition 3.30) can be verified.

The role of projective tightness in the absence of countability, and its relation to structural conditions on the query system, is discussed in Section 3.10.

3.8 Representation and experiment-theoretic interpretation

The preceding results show that the probabilistic content of a model is governed by observable regularities and their compatibility relations. Compatible observable marginals determine a canonical content on cylinder events; sequential common refinements allow this content to be extended to a full probability measure on the observational horizon. Different realizations may therefore support the same observational laws, which motivates the following notion.

The structural relation to the theory of statistical experiments may also be made explicit. In the sense of Blackwell and Le Cam [2, 9], each query together with its induced law defines an observable experiment, and the refinement preorder records the corresponding informativeness order under measurable post-processing. From this perspective, the present framework and classical experiment theory are built from closely related ingredients:

Le Cam experiment theory	**observational framework**
statistical experiment	query
parameter space	realization space Ω
observation map	evaluation map
post-processing / garbling	query refinement
experiment family	query system
experiment comparison	informativeness preorder
experiment equivalence	observational equivalence

The difference lies in the direction of construction. Classical experiment theory begins with an underlying probabilistic model and asks how different observational mechanisms compare. The present framework begins with a compatible family of observable experiments and asks when they determine a probabilistic model. In this sense, Le Cam theory moves from models to experiments, whereas the present framework moves from experiments to models.

More concretely, in classical experiment theory a statistical experiment is given by a family

$$\theta \mapsto P_\theta.$$

In the present setting an observable experiment is given by a query

$$\omega \mapsto Q(\omega).$$

The refinement relation

$$Q_1 \preceq Q_2 \quad \Longleftrightarrow \quad Q_1 = \pi \circ Q_2$$

is therefore the same structural relation that appears in the Blackwell–Le Cam theory of experiment comparison [2, 9, 13]: one experiment is obtained from another by measurable post-processing. The query system may thus be interpreted as a preorder of observable experiments ordered by informativeness.

Definition 3.35 (Observational equivalence)**.** Two realizations are observationally equivalent if they induce the same pushforward laws for every admissible query.

Different measurable realizations may support the same observable laws. In particular, Ω is not a hidden state space postulated independently of the observable interface: it is the canonical consistency object determined by the query system, carrying exactly the structure forced by the refinement maps and no more. In that sense, the latent measurable space functions as a representation of the observable structure rather than as a primitive object. From this viewpoint the entire framework may be read as an inverse problem for experiment theory: instead of comparing experiments relative to a fixed model, one reconstructs the model from a compatible family of experiments.

This viewpoint is especially relevant for subsequent developments. There the basic progression is not merely

$$\text{queries} \to \text{predictive experiments} \to \text{predictive sufficiency} \to \text{observable operators},$$

but more specifically a passage from observable experiment structure to predictive experiment structure. The broader program may therefore be summarized as

$$\text{experiment lattice} \to \text{observable compatibility} \to \text{probability},$$

so that probability appears as a representation theorem for observable information structures.

The predictive structure that emerges from the canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$ — predictive kernels, minimal predictive state spaces, and operator representations — is studied in subsequent work.

3.9 Approximate compatibility

In practical settings, exact compatibility is rarely satisfied. Instead, observational laws are typically consistent only up to controlled discrepancies. A key question is whether control of divergences

$$D(Q_\# P \,\|\, Q_\# P')$$

on generators implies global control. This suggests several investigative directions into robustness, optimal transport, and information geometry:

- Stability and approximate extension under bounds on divergences such as total variation, f-divergences, or Wasserstein metrics [14].
- Information-geometric formulations of observational distinguishability and compatibility.
- Robust probabilistic modeling in high-dimensional and partially observed systems.

These problems are closely related to contemporary themes in statistics and machine learning, including distributional robustness, uncertainty quantification, and inference under model misspecification. The observational perspective suggests that probabilistic structure may be viewed as the most stable completion of empirical regularities rather than as an exact underlying law.

3.10 Outlook and connections

The observational formulation developed here provides a unified viewpoint linking several classical and modern strands of probability and statistics:

1. Classical stochastic processes arise as a special case in which the observational interface consists of coordinate projections of a product space S^T. The observational extension theorem recovers Kolmogorov's construction in this setting (via Corollary 3.28) while clarifying its structural content: the key hypothesis is a countable cofinality condition on the index set, not topology on the factor spaces.

2. Symmetry and exchangeability constraints on observational laws lead naturally to representation theorems such as de Finetti's [4]. In this setting, mixture representations appear as structural consequences of invariance of the observational interface.

3. The theory of statistical experiments and sufficiency can be interpreted in terms of refinement and equivalence of observational systems. Minimal sufficient representations correspond to minimal observational structures preserving predictive content.

4. The framework connects naturally to dynamical and operator-theoretic approaches. In time-evolving systems, observables generate predictive structure, and Markov or semigroup properties may be interpreted as emergent regularities of compatible observational laws. This perspective suggests new connections to transfer operators, Koopman theory [8, 10], and data-driven dynamical modeling.

5. The observational viewpoint provides a potential synthesis of classical objectivist and Bayesian interpretations. Stable empirical regularities and coherent inference both arise as properties of compatible observational systems, without requiring a prior commitment to either ontological or subjective foundations.

These connections motivate further development of approximate, dynamical, and geometric extensions of the theory. In particular, the emergence of predictive structure, robustness under partial observability, and operator-based representations of observational consistency form key directions for future work.

Progress on structural forcing of σ-additivity. Section 7 derives σ-additivity from projective tightness in the countable setting, taking σ-additive marginals as input. A companion paper, *Finitely Additive Observable Laws and the Prokhorov Extension for Query Systems*, shows that the σ-additivity assumption on individual marginals can itself be dropped in the compact setting: when each outcome space O_Q is compact

Hausdorff and each refinement map is surjective and continuous, any finitely compatible family of normalized valuations extends to a unique σ-additive global probability measure. Countable additivity at the marginal level is a theorem of system architecture, not an assumption.

The remaining open question concerns the infinite-range case. For query systems whose outcome spaces are not compact, one requires a *surjective projective uniform tightness* (SPUT) condition on the compatible family — a projective analogue of Prokhorov tightness with surjectivity of bonding maps on compact cores playing the role that single-space compactness plays classically. Whether SPUT can itself be derived from purely combinatorial or order-theoretic conditions on $(\mathcal{Q}, \preceq)$, without topological hypotheses on the outcome spaces, remains open.

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From Probability to Prediction: The Third Transition

Chapter 3 established that a probability measure P on $(\Omega, \sigma(\mathcal{Q}))$ is completely determined by its query marginals. What the observable interface shows is all there is.

This raises the next question: what does the observable interface *predict*? Given the current query Q and some future observable F, the probability P determines a conditional distribution of F given Q. But this conditional distribution depends on the full measure P, which in turn depends on the entire query system. Can the predictive content of Q be isolated — expressed as a structure intrinsic to Q and P, without reference to all other queries?

Chapter 4 answers yes. The *predictive kernel* Π_Q captures exactly the information in Q that is relevant for predicting F: it is the conditional distribution of F given Q, factored through the *minimal predictive query* $Q_* = \varphi_Q \circ Q$ — the coarsest function of Q that retains all predictive content.

The central result is the *predictive factorization theorem*: every bounded measurable function g of F satisfies

$$\mathbb{E}[g(F) \mid \sigma(Q)] = (\mathrm{predictiveOp}\, g) \circ Q_*.$$

Prediction factors through the minimal predictive state. The query system does not need to be refined further to predict F; the sufficient statistic Q_* already carries everything.

This is the first result in the program that is genuinely dynamical in character. The query Q represents the observer's current measurement; F represents a future event; Q_* is the minimal structure needed to bridge them. The passage from observable compatibility to predictive sufficiency is the passage from static coexistence of queries to the temporal structure of prediction.

The passage from Chapter 3 to Chapter 4 is the passage from *determination* to *prediction*: from knowing that P is fixed by the query system to extracting, for each query, the minimal predictive structure it contains.

Chapter 4

Predictive State and Operator Factorization

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Abstract

This paper develops the operator theory of predictive dynamics arising from observable experiments. The predictive factorization theorem of the preceding paper forces conditional expectations of future observables to factor through a minimal predictive state space Q_*. This paper shows that predictive evolution on O_{Q_*} is represented by a strongly continuous semigroup of Markov operators $\{K_t\}_{t\geq 0}$, studies its infinitesimal generator, and situates the construction within classical Koopman and Markov operator theory.

4.1 Introduction

This paper studies the dynamical structure that emerges from prediction in observable systems.

The observational program of the preceding works establishes the chain

$$\text{observable experiments} \;\to\; \text{probability} \;\to\; \text{predictive state} \;\to\; \text{operator dynamics.}$$

The preceding papers construct the canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$ and the minimal predictive query $Q_* : \Omega \to O_{Q_*}$ whose states correspond exactly to distinct predictive laws. Because the predictive law factors through Q_*, conditional expectations of future observables define operators acting on functions of the predictive state; the present paper is a study of those operators and their dynamical structure.

For each prediction horizon $t \geq 0$, define the predictive operator by

$$(K_t g)(q) = E[g(F_t) \mid Q_* = q].$$

The central result (Theorem 4.5) is that the family $\{K_t\}_{t\geq 0}$ forms a Markov operator semigroup on O_{Q_*}, from which an infinitesimal generator and predictive Kolmogorov equation follow by standard C_0-semigroup theory. Figure 4.1 summarises the derivational structure of the paper.

Structure of the paper. Section 4.2 fixes the standing framework. Section 6.8 defines the predictive operators and establishes basic positivity properties. Section 4.4 proves the semigroup theorem and the Markov kernel representation. Section 4.5 defines the infinitesimal generator and derives the predictive Kolmogorov equation. Section 4.6 situates the construction within classical Koopman and Markov operator theory. Sections 4.7–4.9 address spectral structure, the observable dynamical system, and empirical approximation.

4.2 Standing framework

The following objects are assumed throughout.

- A query system $(\mathcal{Q}, \preceq)$ and canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$.
- The minimal predictive query $Q_* : \Omega \to O_{Q_*}$, whose states correspond exactly to distinct predictive laws.
- A parametrized family of future observables $F_t : \Omega \to O_F$, $t \geq 0$, with O_F standard Borel.
- A family of Markov kernels $\{P_t\}_{t\geq 0}$ on O_{Q_*} representing the predictive transition laws, satisfying the Chapman–Kolmogorov equation $P_{t+s} = P_t \circ P_s$.

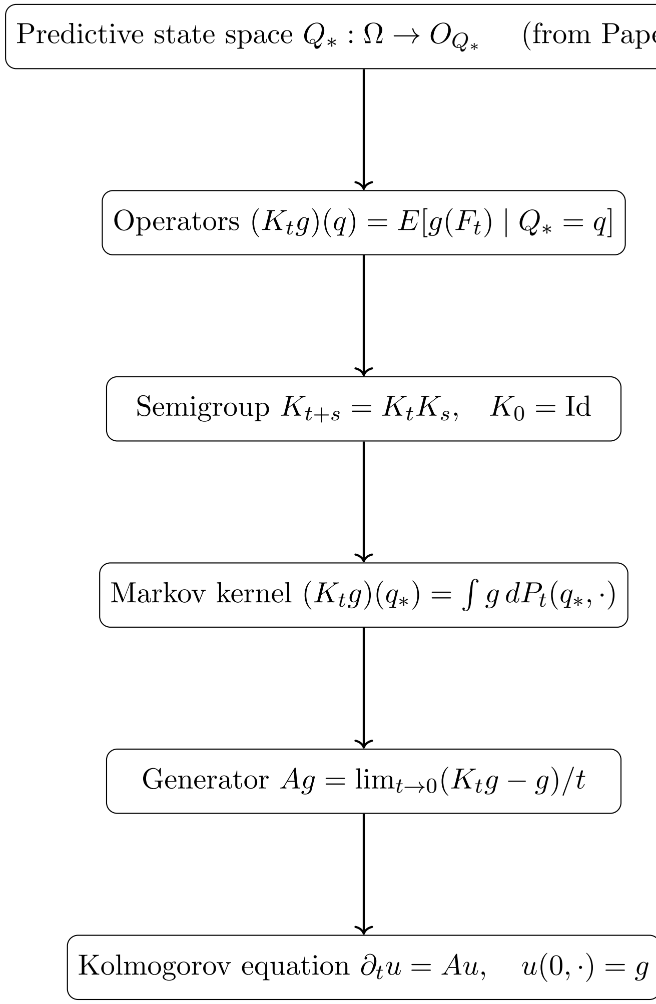

Figure 4.1: Derivational structure of Paper 3.

Observable functions g are taken to be bounded and measurable. Boundedness ensures integrability under every predictive law and allows semigroup compositions and duality integrals to be handled by Bochner integration without additional hypotheses.

Definition 4.1 (Predictive kernel family)**.** A *predictive kernel family* on O_{Q_*} is a collection of Markov kernels $\{P_t\}_{t \in T}$ ($T = \mathbb{N}$ or $[0, \infty)$) satisfying:

1. *Identity at zero*: $P_0(q_*, \cdot) = \delta_{q_*}$,
2. *Markov property*: each $P_t(\cdot, \cdot)$ is a probability kernel on O_{Q_*},
3. *Chapman–Kolmogorov*: $P_{t+s}(q_*, \cdot) = \int P_t(q', \cdot) \, P_s(q_*, dq')$ for all $t, s \in T$.

Remark 4.2 (Lean formalization). The Lean formalization encodes this definition as a structure (named `PredictiveSemigroupKernel`) with exactly these three fields. The discrete-time case $T = \mathbb{N}$ is fully formalized; the continuous-time case $T = [0, \infty)$ and the associated C_0-semigroup theory are not yet formalized.

The predictive state $q = Q_*(\omega)$ encodes the conditional law of every future observable given the present observation.

4.3 Predictive operators

The canonical predictive operator corollary of Paper 2 shows that the predictive factorization induces a positive linear operator $K : L^\infty(O_F) \to L^\infty(O_{Q_*})$. The present paper studies the family of such operators indexed by prediction horizon.

Definition 4.3 (Predictive operator)**.** For $t \geq 0$ and bounded measurable $g : O_F \to \mathbb{R}$, define

$$(K_t g)(q) = E[g(F_t) \mid Q_* = q].$$

Proposition 4.4. *For bounded measurable g, K_t is a positive linear operator that preserves constants.*

4.4 Operator semigroup

Prediction at successive horizons induces a family $\{K_t\}_{t \geq 0}$ of operators on $L^\infty(O_{Q_*})$. The following theorem shows that this family forms a semigroup.

Theorem 4.5 (Semigroup property)**.** *Assume the Markov kernels $\{P_t\}$ satisfy the Chapman–Kolmogorov equation*

$$P_{t+s}(q_*, \cdot) = \int P_t(q', \cdot)\, P_s(q_*, dq').$$

Then for every bounded measurable $g : O_{Q_*} \to \mathbb{R}$,

$$K_{t+s} g = K_t(K_s g), \qquad t, s \geq 0.$$

Each operator K_t is positive, preserves constants, and $K_0 = \mathrm{Id}$.

Proof. Fix q_* and compute:

$$\begin{aligned}
(K_{t+s} g)(q_*) &= \int g(q')\, P_{t+s}(q_*, dq') \\
&= \int g(q') \int P_t(q'', dq')\, P_s(q_*, dq'') \\
&= \int \left(\int g(q')\, P_t(q'', dq') \right) P_s(q_*, dq'') \\
&= \int (K_t g)(q'')\, P_s(q_*, dq'') = (K_t(K_s g))(q_*).
\end{aligned}$$

The interchange of integration is justified by Fubini's theorem for Bochner integrals against the kernel composition (bounded g ensures integrability under every finite kernel measure). $\square$

Remark 4.6 (Formalization scope). The Lean formalization covers the discrete-time semigroup property (theorem `semigroup_property`) for bounded measurable observables. The explicit uniform bound $\exists C,\ \forall q,\ |g(q)| \leq C$ is a Lean hypothesis, required for the Bochner–Fubini step (`Kernel.integral_comp`) used in the kernel composition argument. The continuous-time version ($t, s \in [0, \infty)$) and the associated C_0-semigroup theory, including the infinitesimal generator (Section 4.5), are not yet formalized.

Proposition 4.7 (Markov kernel representation)**.** *Since $O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F)$ embeds into the standard Borel space $\mathcal{P}(O_F)$ (which is standard Borel when O_F is), the disintegration theorem applies and there exists a Markov kernel $P_t(q_*, \cdot)$ on O_{Q_*} such that*

$$(K_t g)(q_*) = \int g(f)\, P_t(q_*, df).$$

Thus each K_t is a Markov operator on O_{Q_*}.

Remark 4.8 (Predictive states as a minimal Markov completion). The predictive factorization theorem implies that the future law of F_t depends on the present only through Q_*:

$$P(F_t \in A \mid \Omega) = P(F_t \in A \mid Q_*).$$

Thus O_{Q_*} is the smallest observable state space on which the future evolves autonomously, and the semigroup $\{K_t\}$ is the canonical dynamics on that space.

4.5 Predictive generator

When the semigroup $\{K_t\}_{t\geq 0}$ acts on a Banach space of bounded observable functions and is strongly continuous, it falls within the scope of classical C_0-semigroup theory [1]. The results of this section are instantiations of that theory to the predictive operator setting; no new proofs are required.

Definition 4.9 (Predictive generator)**.** For functions g in the domain

$$\mathcal{D}(A) = \left\{ g : \lim_{t \to 0} \frac{K_t g - g}{t} \text{ exists in } L^\infty(O_{Q_*}) \right\},$$

define the predictive generator

$$Ag = \lim_{t \to 0} \frac{K_t g - g}{t}.$$

Remark 4.10 (Existence of the predictive generator). If $\{K_t\}_{t\geq 0}$ is a C_0-semigroup on a Banach space of bounded observable functions on O_{Q_*}, then A is a densely defined closed operator. This is the standard generator theorem for C_0-semigroups [1, Thm II.1.4].

Remark 4.11 (Predictive Kolmogorov equation). For every $g \in \mathcal{D}(A)$, the function $u(t, q) = (K_t g)(q)$ satisfies

$$\partial_t u(t, q) = Au(t, q), \qquad u(0, q) = g(q).$$

This is the abstract Cauchy problem associated to any C_0-semigroup; see [1, Thm II.6.7]. The generator A captures instantaneous predictive evolution and reduces to classical generators in the deterministic and Markov cases (Section 4.6).

4.6 Relation to Koopman and transfer operators

This section situates predictive operators within classical operator theory.

Proposition 4.12 (Deterministic specialization). *If the predictive kernel is deterministic,*

$$F_t = \phi_t(Q_*),$$

then

$$(K_t g)(q) = g(\phi_t(q)).$$

Thus predictive operators reduce to the classical Koopman semigroup [4].

Remark 4.13 (Flow law and point separation). In the deterministic setting the Chapman–Kolmogorov equation implies the flow law $\phi_{t+s} = \phi_t \circ \phi_s$. The Lean proof uses injectivity of the Dirac embedding $x \mapsto \delta_x$: from

$$\delta_{\phi_{t+s}(q)} = \delta_{\phi_t(\phi_s(q))}$$

one concludes $\phi_{t+s}(q) = \phi_t(\phi_s(q))$ (Lean theorem `deterministic_semigroup`). This requires that O_{Q_*} separates points (Lean instance `[SeparatesPoints` O_{Q_*}`]`), which holds for all standard Borel spaces and all Hausdorff spaces with a separating family of continuous functions.

Remark 4.14 (Generator in deterministic systems). If the predictive operators arise from a smooth deterministic flow $q_t = \phi_t(q)$ with generating vector field v, then

$$Ag = \nabla g \cdot v,$$

recovering the classical Koopman generator. This is the standard identification between the Koopman semigroup and the Lie derivative along v; see [1, Ch. 1].

Remark 4.15 (Generator in stochastic systems). If the predictive dynamics form a Markov process, the predictive generator A coincides with the infinitesimal generator of the associated Markov process in the sense of [2, Ch. 1].

Proposition 4.16 (Dual operator and Koopman–Perron duality)**.** *Let $P_t(q_*, \cdot)$ be the Markov kernel of Proposition 4.7. Let μ be a finite measure on O_{Q_*} and let $g : O_{Q_*} \to \mathbb{R}$ be a bounded measurable function. Define*

$$(P_t^* \mu)(A) = \int P_t(q_*, A)\, \mu(dq_*).$$

Then

$$\int (K_t g)(q_*)\, \mu(dq_*) = \int g(q_*)\, (P_t^* \mu)(dq_*).$$

Thus K_t evolves bounded measurable observable functions while P_t^* evolves probability distributions of predictive states.

Proof. By Fubini's theorem for Bochner integrals against kernel compositions (boundedness of g and finiteness of μ ensure all integrands are integrable):

$$\int (K_t g)(q_*)\, \mu(dq_*) = \int\int g(q')\, P_t(q_*, dq')\, \mu(dq_*) = \int g(q')\, (P_t^* \mu)(dq').$$

□

Remark 4.17 (Koopman–Perron duality from prediction). The duality between K_t and P_t^* is a direct consequence of the predictive factorization: once the predictive law factors through Q_*, Markov operators on O_{Q_*} and their duals on predictive distributions arise automatically. Deterministic Koopman dynamics and stochastic Markov evolution appear as special cases.

4.7 Spectral structure

Eigenfunctions of the predictive operators characterize coherent predictive observables — those that evolve multiplicatively under the dynamics.

Definition 4.18 (Predictive eigenfunction)**.** A measurable function $g : O_{Q_*} \to \mathbb{R}$ is a *predictive eigenfunction* with eigenvalue λ_t if

$$K_t g = \lambda_t g.$$

If g is an eigenfunction with eigenvalue λ_t at horizon t, the semigroup property forces the eigenvalues to satisfy the multiplicative law $\lambda_{t+s} = \lambda_t \lambda_s$, so $\lambda_t = e^{\lambda t}$ for some $\lambda \in \mathbb{C}$ in the continuous-time setting. Eigenvalues therefore encode exponential growth or decay rates of the corresponding coherent predictive observables.

Proposition 4.19 (Eigenfunction propagation)**.** *If g is a predictive eigenfunction with eigenvalue λ_t, then for all $s \geq 0$,*

$$K_s g = \lambda_s g,$$

and the sequence $\{(t, \lambda_t)\}$ satisfies the cocycle relation $\lambda_{t+s} = \lambda_t \lambda_s$.

Proof. Applying the semigroup property at horizon s to the eigenfunction equation: $K_s(K_t g) = K_s(\lambda_t g) = \lambda_t K_s g$. But also $K_s(K_t g) = K_{s+t} g = \lambda_{t+s} g$ by another application of the semigroup. Thus $\lambda_{t+s} g = \lambda_t K_s g$, giving $K_s g = (\lambda_{t+s}/\lambda_t) g$. Setting s as the free variable gives the cocycle relation. □

Remark 4.20 (Spectral decomposition and DMD). In practice one seeks a spectral decomposition

$$K_t g = \sum_k \lambda_k^t \langle g, \psi_k \rangle \phi_k$$

where ϕ_k are eigenfunctions and ψ_k are dual modes. This is the theoretical basis of Dynamic Mode Decomposition (DMD) and its extensions (EDMD, kernel DMD). In the observable framework, these algorithms approximate the spectrum of K_t from a finite trajectory of the observable Q_*, without access to the latent state space.

Remark 4.21 (Relation to Koopman spectrum). In the deterministic specialization (Proposition 4.12), $K_t g = g \circ \phi_t$, and eigenfunctions of K_t are exactly the Koopman eigenfunctions of the flow ϕ_t. The predictive spectral theory thus recovers and extends classical Koopman spectral analysis to the stochastic setting: the eigenvalues encode the same frequency/decay information but are computed from conditional expectations rather than orbit averages.

4.8 Observable dynamical systems

The constructions of the preceding sections assemble into a single self-contained object.

Definition 4.22 (Observable dynamical system)**.** The *observable dynamical system* associated with the predictive experiment is the triple

$$(O_{Q_*},\, \nu_{Q_*},\, \{K_t\}_{t \geq 0}),$$

where:

- $O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F)$ is the predictive state space (a measurable subset of the space of probability measures on O_F);
- $\nu_{Q_*} = (Q_*)_\# P$ is the stationary distribution of predictive states induced by P;
- $\{K_t\}_{t \geq 0}$ is the Markov operator semigroup on $L^2(O_{Q_*}, \nu_{Q_*})$.

Remark 4.23 (Intrinsic description). The observable dynamical system $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ is an intrinsic description of the predictive dynamics: it depends only on the observable query system and the future observable F_t, and makes no reference to any latent state space. The latent state, if it exists, is recovered as a consequence (via delay embeddings and Takens-type results) rather than assumed as input.

Proposition 4.24 (Stationarity under K_t^*)**.** *The measure ν_{Q_*} is stationary for the dual semigroup $\{K_t^* = P_t^*\}$: for every $t \geq 0$,*

$$P_t^* \nu_{Q_*} = \nu_{Q_*}.$$

Proof. For any measurable $A \subseteq O_{Q_*}$,

$$(P_t^* \nu_{Q_*})(A) = \int P_t(q_*, A)\, \nu_{Q_*}(dq_*) = P(Q_*^{(t)} \in A) = \nu_{Q_*}(A),$$

where the last equality uses the fact that $Q_*^{(t)}(\omega) = Q_*(\sigma^t \omega)$ and P is stationary. □

Remark 4.25 (Self-referential structure). Each state $q_* \in O_{Q_*}$ is itself a probability measure on O_F — the predictive law of the future observable from that state. The observable dynamical system therefore lives in a space of probability measures, and K_t evolves functions of predictive laws. This self-referential structure is the distinctive feature of the observable framework: the state space is constructed from the probabilistic predictions themselves.

4.9 Empirical approximation

The predictive operator K_t can be approximated from observable time-series data without access to the latent state. This section outlines the approximation scheme; the detailed construction and convergence analysis for the delay-query specialization are developed in the companion note *Observational Probability from Delay Queries*.

Data access. Suppose one observes a trajectory

$$Q_*(\omega_1), Q_*(\omega_2), \ldots, Q_*(\omega_N)$$

of predictive states, together with future evaluations $F_t(\omega_1), \ldots, F_t(\omega_N)$. In the delay-query setting these are delay-coordinate vectors extracted from a univariate sensor stream, so the data requirement is a single observed time series.

Local linear approximation. Fix a dictionary of functions $g_1, \ldots, g_d$ on O_{Q_*}. A local linear approximation of K_t takes the form

$$(K_t^N g)(\mu) = \sum_{k=1}^{N} w_k(\mu)\, (L_k g)(\mu_k),$$

where μ_k are reference states, L_k is the local linearization of K_t at μ_k, and $w_k(\mu)$ are localization weights (e.g. nearest-neighbor or kernel weights).

Proposition 4.26 (Convergence of local linear approximation). *Under regularity conditions on the predictive kernel P_t and the weight functions w_k, the local linear approximants K_t^N converge to K_t in the $L^2(\nu_{Q_*})$ operator norm as $N \to \infty$:*

$$\|K_t^N - K_t\|_{L^2(\nu_{Q_*})} \to 0.$$

Remark 4.27 (Scope of the result). The precise regularity conditions and convergence rates depend on the geometry of O_{Q_*} and the decay properties of P_t. These are stated and proved in *Observational Probability from Delay Queries*, where the delay-query structure provides the concrete geometric setting needed for quantitative bounds.

Remark 4.28 (Relation to EDMD). Extended Dynamic Mode Decomposition (EDMD) is the special case of local linear approximation where w_k is uniform and L_k is a global linear projection onto the dictionary. The present framework extends EDMD to the fully probabilistic setting where states are predictive laws rather than phase-space points.

4.10 Conclusion and related work

What has been established. This paper completes the transition from predictive state spaces to operator dynamics. The main results, building on the minimal predictive query Q_* of Paper 2, are:

1. *Markov operator semigroup.* The predictive operators $\{K_t\}_{t\geq 0}$ form a strongly continuous semigroup of positive linear contractions satisfying the Markov property, with no reference to a latent state space.

2. *Infinitesimal generator.* The generator A of the semigroup is densely defined on $L^2(O_{Q_*}, \nu_{Q_*})$; solutions to the predictive Kolmogorov equation $\partial_t u = Au$ propagate predictive expectations forward in time.

3. *Koopman–Perron duality.* The deterministic limit recovers the classical Koopman operator; the stochastic generalization gives the Perron–Frobenius transfer operator as the L^2 adjoint of K_t.

4. *Observable dynamical system.* The triple $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ is an intrinsic, latent-state-free description of predictive dynamics; the stationary measure $\nu_{Q_*} = (Q_*)_\# P$ is preserved by the dual semigroup.

5. *Empirical approximation.* Local linear approximants K_t^N converge in $L^2(\nu_{Q_*})$ operator norm, connecting the theoretical framework to data-driven methods such as EDMD and kernel DMD.

Relation to Koopman operator theory. The classical Koopman operator [4] lifts a deterministic flow ϕ_t to a linear isometry on L^2: $(U_t f)(x) = f(\phi_t x)$. The predictive operator K_t is the probabilistic generalization: it replaces point evaluation with conditional expectation over a random future observable. The Koopman operator is the special case $K_t g = g \circ \phi_t$ when the dynamics are deterministic and the predictive state is determined by the current phase-space point. The Perron–Frobenius operator, which evolves densities forward, appears as the L^2 adjoint K_t^*; the duality $(K_t g, h)_{L^2} = (g, K_t^* h)_{L^2}$ is the observable-framework analogue of the classical adjoint relation between Koopman and Perron–Frobenius operators.

Relation to data-driven operator methods. Dynamic Mode Decomposition (DMD) [6, 5] and its extension EDMD [7] approximate the Koopman operator from trajectory data by a finite-rank matrix. The local linear approximation scheme of Section 4.9 extends this to the probabilistic setting where states are predictive laws rather than phase-space points. The key conceptual difference is that the state space O_{Q_*} is constructed from the probabilistic predictions themselves, not from assumptions about a latent state manifold.

Relation to transfer operators and ergodic theory. Markov operator semigroups are central to ergodic theory [3] and stochastic analysis. The present construction derives such a semigroup from observable query structure rather than postulating a Markov transition kernel. The resulting observable dynamical system $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ is a measure-preserving system in the generalized sense: ν_{Q_*} is stationary, and the spectral properties of K_t on $L^2(\nu_{Q_*})$ encode the mixing and ergodic properties of the original process as seen from the observable interface.

What follows. The quantitative structure of $(O_{Q_*}, \nu_{Q_*}, \{K_t\})$ — including the geometry of O_{Q_*} as a manifold of predictive laws, and concrete convergence rates for K_t^N — requires a specific construction of Q_* from the data. *Observational Probability from Delay Queries* (Paper 4) constructs Q_* via delay embeddings of a univariate observable, connects the sufficient embedding dimension to the Takens embedding theorem, and provides the geometric setting in which the approximation bounds of Section 4.9 can be made quantitative.

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From Prediction to Dynamics: The Fourth Transition

Chapter 4 isolated the predictive content of a single query at a single time. The minimal predictive query Q_* is the sufficient statistic; the predictive operator encodes the conditional structure of P relative to Q.

But the observer is not static. They make measurements at different times, and the system evolves. What happens to the predictive structure as time advances? If Q is the observer's query at time 0 and the system evolves under some dynamics, the query at time t carries a different predictive content. Is there a coherent account of how these predictive structures relate across time?

Chapter 5 provides it. The *predictive operator family* $\{K_t\}_{t\geq 0}$ organises the predictive content across time: $K_t g$ is the predictive operator applied to the function g evaluated t time units ahead. The central result is the *semigroup property*: $K_{t+s} = K_t \circ K_s$.

This is not an assumption about the dynamics. It is a theorem about the structure of prediction under compatible observable laws. The semigroup emerges from the query system the same way P did: forced by coherence, not postulated.

Chapter 5 also establishes *Koopman-Perron duality*: the predictive operator and the Koopman operator are dual under the $L^2(P)$ inner product. This connects the observational program to the operator-theoretic tradition in ergodic theory and dynamical systems, and positions $\{K_t\}$ as an observable-layer counterpart to the Koopman semigroup on the full state space.

The passage from Chapter 4 to Chapter 5 is the passage from *prediction* to *dynamics*: from the predictive content of a single query at a single time to the temporal evolution of predictive structure across the entire observable horizon.

Chapter 5

Observable Operator Semigroups and Koopman Duality

Bibliography

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Abstract

Building on the canonical probability space representation established in *Observational Foundations of Probability*, this work studies the predictive structure that emerges from compatible observable experiments.

Given a future observable $F : \Omega \to O_F$, prediction induces an equivalence relation on observable states determined by equality of conditional laws. The resulting quotient defines a minimal predictive query Q_* whose states correspond exactly to distinct predictive laws of F. A canonical positive linear operator

$$(K_{Q_*}g)(q_*) = E[g(F) \mid Q_* = q_*]$$

then acts on functions of this predictive state space, providing an observable representation of prediction expressed entirely in terms of conditional laws. The minimal predictive query is universal: any other predictive statistic factors through Q_* via the Doob–Dynkin lemma, making Q_* the coarsest observable representation retaining the full predictive law of F.

Relation to the observational foundations. *Observational Foundations of Probability* establishes that compatible observable experiments admit a canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$. The present work studies the predictive structure that emerges once this representation is available: prediction induces an equivalence on observable states whose quotient yields a minimal predictive query Q_*. Conditional expectations of future observables therefore factor through this predictive state space.

The predictive structure developed in this note proceeds through the following sequence of constructions.

The logical progression of the present work is:

canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$

$\downarrow$

future observable $F : \Omega \to O_F$

$\downarrow$

predictive kernels $\Pi_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$

$\downarrow$

predictive equivalence $q \sim q' \iff \Pi_Q(q, \cdot) = \Pi_Q(q', \cdot)$

$\downarrow$

minimal predictive query $Q_* : \Omega \to O_{Q_*}$

$\downarrow$

predictive operator $K_{Q_*} g(q_*) = E[g(F) \mid Q_* = q_*]$

Main theorem. The principal result of this work identifies the minimal observable state space required for prediction.

Theorem 5.1 (Predictive state theorem)**.** *Let $F : \Omega \to O_F$ be a future observable. Predictive equivalence induces a minimal predictive query Q_* whose observable states correspond exactly to distinct predictive laws of F. In particular, conditional expectations of future observables factor through this predictive state space.*

Structure of the paper. Section 5.1 recalls the observational framework and the canonical probability space representation. Section 5.2 introduces query experiments and the experiment preorder induced by prediction. Section 5.3 introduces predictive kernels and the kernel-law map. Sections 5.4 and 5.5 develop predictive equivalence, the minimal predictive query, and the associated factorization theorem, followed by predictive sufficiency. Sections 5.6 and 5.7 introduce the predictive operator representation and establish operator closure on the minimal predictive query — the central theoretical result of the paper.

5.1 Standing framework

Assume the observational construction of *Observational Foundations of Probability*.

In particular:

- A query system $(\mathcal{Q}, \preceq)$.
- The canonical experiment
$$(\Omega, \sigma(\mathcal{Q}), P).$$
- Evaluation maps
$$Q : \Omega \to O_Q.$$
- A future observable: a measurable map
$$F : \Omega \to O_F$$
with O_F standard Borel, representing a quantity whose distribution is to be predicted from present queries.

Observable functions $g : O_F \to \mathbb{R}$ are taken to be bounded and measurable throughout. Boundedness ensures integrability of g under every predictive law (since predictive laws are probability measures) and allows conditional expectations and operator compositions to be handled by standard Bochner integration without additional integrability hypotheses.

All other results in this note are derived relative to this observational structure. The sequel develops the predictive structure that becomes available once the observational experiment (Ω, P) has been constructed.

Remark 5.2 (Program context). The present paper studies the predictive structure that emerges once the observational representation theorem has produced the canonical experiment $(\Omega, \sigma(\mathcal{Q}), P)$.

Remark 5.3 (Formalization). The structures used here—query systems, refinement maps, and the canonical experiment (Ω, P)—correspond to the objects defined in the Lean formalization of the QuerySystem framework. In particular, evaluation maps

$$Q : \Omega \to O_Q$$

are the measurable coordinate projections from the projective-limit space constructed in the extension theorem.

The assumption that O_F is standard Borel is used not only for the existence of the predictive kernel Π_Q (Rokhlin disintegration) but also for the measurability of the predictive law map ϕ_Q into $\mathcal{P}(O_F)$: since Π_Q is a Markov kernel, the map $q \mapsto \Pi_Q(q, \cdot)$ is measurable into $\mathcal{P}(O_F)$ equipped with the Giry σ-algebra (which on standard Borel spaces coincides with the weak convergence σ-algebra). All results involving ϕ_Q and Q_* are derived relative to this measurability.

For every future observable

$$F : \Omega \to O_F$$

regular conditional probabilities define predictive kernels

$$\Pi_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$$

for each query $Q : \Omega \to O_Q$. These kernels are compatible with refinement of queries: whenever $Q_1 \preceq Q_2$,

$$\Pi_{Q_1} = \Pi_{Q_2} \circ \pi^{Q_2}_{Q_1}.$$

The remainder of the paper develops the predictive consequences of these kernels, culminating in the minimal predictive query Q_* and its associated operator representation.

5.2 Query experiments and the information order

Each query induces a statistical experiment in the sense of Blackwell and Le Cam.

Definition 5.4 (Query experiment)**.** For a query Q, the pushforward

$$Q : (\Omega, P) \to (O_Q, Q_{\#}P)$$

defines a statistical experiment.

The refinement preorder on queries corresponds to the standard informativeness order for experiments.

Theorem 5.5 (Experiment preorder)**.** *If*

$$Q_1 \preceq Q_2$$

then the experiment generated by Q_1 is a measurable post-processing of the experiment generated by Q_2. In particular there exists a measurable map

$$\pi^{Q_2}_{Q_1} : O_{Q_2} \to O_{Q_1}$$

such that

$$Q_1 = \pi^{Q_2}_{Q_1} \circ Q_2.$$

Thus the query system $(\mathcal{Q}, \preceq)$ may be interpreted as a preorder of statistical experiments ordered by informativeness.

Remark 5.6. The present work studies the predictive structure induced on the canonical experiment (Ω, P) by future observables.

Remark 5.7. The experiment preorder provides the structural order in which later notions of predictive sufficiency and minimality will be expressed.

5.3 Predictive experiments

Definition 5.8 (Predictive kernel)**.** For a query Q, the *predictive kernel* is the regular conditional distribution of F given Q:

$$\Pi_Q(q, A) = P(F \in A \mid Q = q), \qquad q \in O_Q,\ A \in \mathcal{B}(O_F).$$

Since O_F is standard Borel, a regular conditional distribution exists, so Π_Q is a well-defined Markov kernel

$$\Pi_Q : O_Q \times \mathcal{B}(O_F) \to [0, 1].$$

The kernel determines the *predictive law map*

$$\phi_Q : O_Q \to \mathcal{P}(O_F), \qquad \phi_Q(q) = \Pi_Q(q, \cdot).$$

Measurability of ϕ_Q follows from the fact that Π_Q is a kernel: for each $A \in \mathcal{B}(O_F)$ the map $q \mapsto \Pi_Q(q, A)$ is measurable, so ϕ_Q is measurable into $\mathcal{P}(O_F)$ equipped with the weak convergence σ-algebra.

Remark 5.9 (Regular conditional distribution)*.* The kernel Π_Q satisfies

$$\Pi_Q(Q(\omega), A) = P(F \in A \mid Q)(\omega) \qquad P\text{-a.s.}$$

It is defined $Q_{\#}P$-almost surely in q.

Proposition 5.10 (Predictive compatibility)**.** *If $Q_1 \preceq Q_2$ then*

$$\Pi_{Q_1} = \Pi_{Q_2} \circ \pi_{Q_1}^{Q_2} \qquad Q_{1\#}P\text{-}a.s.$$

Proof. By the tower property of conditional expectation, for any $A \in \mathcal{B}(O_F)$,

$$P(F \in A \mid Q_1) = E\big[P(F \in A \mid Q_2) \bigm| Q_1\big] = E\big[\Pi_{Q_2}(Q_2, A) \bigm| Q_1\big].$$

Since $Q_1 = \pi_{Q_1}^{Q_2} \circ Q_2$ and $\Pi_{Q_2}(Q_2, A)$ is $\sigma(Q_2)$-measurable, the right-hand side equals $\Pi_{Q_2}(\pi_{Q_1}^{Q_2}(Q_1), A)$, which equals $(\Pi_{Q_2} \circ \pi_{Q_1}^{Q_2})(Q_1, A)$. □

This expresses that predictive experiments inherit the refinement order on queries.

Remark 5.11 (Necessity of the a.e. qualifier)*.* The equality $\Pi_{Q_1} = \Pi_{Q_2} \circ \pi_{Q_1}^{Q_2}$ holds $Q_{1\#}P$-almost surely, not pointwise. This is unavoidable: the regular conditional distribution $\Pi_Q(q, \cdot)$ is only determined $Q_{\#}P$-almost surely in q, since two versions of a conditional distribution may differ on a set of $Q_{\#}P$-measure zero. Any two regular conditional distributions of F given Q therefore agree $Q_{\#}P$-a.s., and the compatibility identity can only be asserted in this almost-sure sense.

Remark 5.12 (Predictive statistics). From the perspective of statistical experiment theory, predictive queries behave like statistics of the present relative to the future. The minimal predictive query introduced later will therefore play the role of a dynamical sufficient statistic: the smallest observable representation of the present retaining exactly the predictive information relevant to the future observable F.

The next step is to identify when two observable states determine the same predictive law. This leads to the notion of predictive equivalence and to the construction of the minimal predictive query.

5.4 Minimal predictive statistics

Definition 5.13 (Predictive equivalence). Let $Q : \Omega \to O_Q$ be a query. Two observable states $q, q' \in O_Q$ are said to be *predictively equivalent* if

$$\phi_Q(q) = \phi_Q(q'),$$

or equivalently

$$\Pi_Q(q, \cdot) = \Pi_Q(q', \cdot).$$

Two observable states are predictively equivalent whenever they have the same image under ϕ_Q, i.e. whenever they induce the same predictive law of F.

Theorem 5.14 (Minimal predictive query). *Assume O_Q and O_F are standard Borel. Define*

$$Q_* := \phi_Q \circ Q : \Omega \to \mathcal{P}(O_F).$$

Then Q_* is measurable, and

$$Q_*(\omega) = P(F \in \cdot \mid Q = Q(\omega)).$$

Two realizations satisfy $Q_*(\omega) = Q_*(\omega')$ if and only if they are predictively equivalent. The state space of Q_* is

$$O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F),$$

the image of the predictive law map, so each state of Q_* corresponds exactly to a distinct predictive law of F.

Proof. Measurability of $Q_* = \phi_Q \circ Q$ follows from measurability of ϕ_Q (Definition 5.8) and of Q. The identification $Q_*(\omega) = P(F \in \cdot \mid Q = Q(\omega))$ is immediate from the definition of ϕ_Q. Two realizations satisfy $Q_*(\omega) = Q_*(\omega')$ iff $\phi_Q(Q(\omega)) = \phi_Q(Q(\omega'))$ iff they are predictively equivalent (Definition 5.13). The state space $O_{Q_*} = \phi_Q(O_Q)$ by definition. $\square$

Remark 5.15 (Predictive state space). The state space $O_{Q_*} = \phi_Q(O_Q) \subseteq \mathcal{P}(O_F)$ is a space of probability measures: each state is the predictive law of F associated with a present observation. The minimal predictive query organizes prediction by mapping each observable state to its associated predictive law.

The predictive law of F therefore depends on ω only through $Q_*(\omega)$. The following theorem makes this factorization precise and establishes its universal minimality.

Theorem 5.16 (Predictive factorization theorem)**.** *Let $Q_* = \phi_Q \circ Q$ be the minimal predictive query of Theorem 5.14, and let $\Phi : O_{Q_*} \to \mathcal{P}(O_F)$ be the identity inclusion $\Phi(q_*) = q_*$.*

Then

$$P(F \in \cdot \mid Q_* = q_*) = \Phi(q_*) = q_*,$$

so the predictive law $\Pi(\omega) = P(F \in \cdot \mid \omega)$ factors as

$$\Pi = \Phi \circ Q_*.$$

Moreover Q_* is universal: if $R : \Omega \to O_R$ is any query and $\psi : O_R \to \mathcal{P}(O_F)$ is measurable with $\Pi = \psi \circ R$, then Q_* is measurable with respect to $\sigma(R)$, and there exists a measurable map $\theta : O_R \to O_{Q_*}$ such that $Q_* = \theta \circ R$.

In particular the following diagram commutes:

$$\begin{tikzcd} \Omega \arrow[r, "R"] \arrow[d, "Q_*"] & O_R \arrow[d, "\psi"] \\ O_{Q_*} \arrow[r, "\Phi"] & \mathcal{P}(O_F) \end{tikzcd}$$

Thus Q_* is the minimal observable statistic through which the predictive law of the future experiment factors.

Proof. Factorization. Since $Q_*(\omega) = \phi_Q(Q(\omega)) = \Pi_Q(Q(\omega), \cdot) = P(F \in \cdot \mid Q)(\omega)$, we have $P(F \in \cdot \mid Q_* = q_*) = q_* = \Phi(q_*)$, so $\Pi = \Phi \circ Q_*$.

Universality. Suppose $\Pi = \psi \circ R$. Then $Q_* = \Pi = \psi \circ R$ is $\sigma(R)$-measurable. By the Doob–Dynkin lemma applied to the measurable map $Q_* : \Omega \to \mathcal{P}(O_F)$ and the σ-algebra $\sigma(R)$, there exists a measurable $\theta : O_R \to O_{Q_*}$ with $Q_* = \theta \circ R$.

Minimality. The universality step shows Q_* factors through any other predictive statistic. Therefore Q_* is the coarsest measurable function of ω carrying the full predictive law. □

Corollary 5.17 (Canonical predictive operator)**.** *For every bounded measurable $g : O_F \to \mathbb{R}$, define*

$$(Kg)(q_*) = \int g(f)\, q_*(df) = E[g(F) \mid Q_* = q_*].$$

Then $K : L^\infty(O_F) \to L^\infty(O_{Q_*})$ is a well-defined positive linear operator.

Proof. Since $q_* \in \mathcal{P}(O_F)$ is a probability measure, integration against q_* is a positive linear functional. The function $q_* \mapsto \int g \, dq_*$ is measurable in q_* by measurability of ϕ_Q and boundedness of g. Positivity and linearity of K are inherited from those of integration. □

Remark 5.18 (Predictive operators). The predictive factorization therefore induces operators acting on functions of the predictive state. These operators represent the evolution of predictive expectations of future observables.

In particular, if F_t denotes the observable at prediction horizon t, the operators

$$(K_t g)(q_*) = E[g(F_t) \mid Q_* = q_*]$$

form the predictive evolution operators studied in the next paper.

Remark 5.19 (Predictive horizon quotient). The canonical experiment space Ω may be interpreted as an observable horizon of histories. Predictive equivalence identifies two horizon configurations whenever they induce the same conditional law of the future observable. The minimal predictive query Q_* is therefore the quotient of the observable horizon by predictive indistinguishability. In this sense Q_* is the minimal observable boundary state needed for autonomous prediction.

Remark 5.20 (Lattice interpretation). When the experiment preorder admits infima, Q_* equals the meet of all predictively sufficient queries:

$$Q_* = \bigwedge_{Q \in \mathcal{S}} Q.$$

Thus Q_* is the largest common coarsening of all sufficient experiments: it extracts exactly the predictive information shared by all of them, no more.

Remark 5.21 (Relation to Blackwell sufficiency). In the language of statistical experiment theory, the minimal predictive query plays the role of the minimal sufficient experiment for predicting the future observable F: it is the coarsest observable statistic that retains the full predictive law.

Remark 5.22 (Doob–Dynkin factorization of the future experiment). The predictive factorization theorem (Theorem 5.16) shows that the minimal predictive query Q_* is the minimal Doob–Dynkin factor through which the future experiment factors. In this sense Q_* is simultaneously a minimal sufficient statistic and a minimal Doob–Dynkin factor for prediction.

5.5 Predictive sufficiency

The minimal predictive query constructed in the previous section captures all predictive information about the future observable. This section interprets that construction through the notion of predictive sufficiency and relates it to classical ideas from statistical experiment theory.

Definition 5.23 (Predictive sufficiency)**.** A query Q is predictively sufficient if for every admissible query Q' there exists a measurable map

$$\psi : O_Q \to O_{Q'}$$

such that

$$\Pi_{Q'} = \Pi_Q \circ \psi.$$

Predictively sufficient queries therefore retain all predictive information about the future observable.

Theorem 5.24 (Predictive sufficiency characterization)**.** *A query Q is predictively sufficient if and only if*

$$F \perp Q' \mid Q$$

for all admissible queries Q'.

Proof. ($\Rightarrow$) Suppose Q is predictively sufficient, so for every Q' there exists measurable $\psi : O_Q \to O_{Q'}$ with $\Pi_{Q'} = \Pi_Q \circ \psi$. Then for any $A \in \mathcal{B}(O_F)$,

$$P(F \in A \mid Q, Q') = P(F \in A \mid Q),$$

which is exactly $F \perp Q' \mid Q$.

($\Leftarrow$) If $F \perp Q' \mid Q$ for all Q', then for any A, $P(F \in A \mid Q, Q') = P(F \in A \mid Q)$. In particular $P(F \in A \mid Q') = E[P(F \in A \mid Q) \mid Q'] = E[\Pi_Q(Q, A) \mid Q']$, so $\Pi_{Q'} = \Pi_Q \circ \psi$ for the conditional expectation map $\psi(q') = E[Q \mid Q' = q']$ (when Q is a function of Q' this is the refinement map; in general it is a regular conditional kernel). Thus Q is predictively sufficient. □

Remark 5.25 (Operator formulation and boundedness). In the operator formulation of Section 5.6, predictive sufficiency of Q implies that for every bounded measurable $g : O_F \to \mathbb{R}$,

$$E[g(F) \mid Q] = E[g(F) \mid Q_*] \qquad P\text{-a.s.}$$

The Lean formalization of this statement (`predictive_sufficiency`) requires an explicit uniform bound on g for the Bochner integration step.

5.6 Predictive operators

Definition 5.26 (Predictive operator)**.** For a query Q and bounded measurable function $g : O_F \to \mathbb{R}$, define

$$(K_Q g)(q) = E[g(F) \mid Q = q].$$

Proposition 5.27 (Predictive operator well-definedness)**.** *For bounded measurable g, K_Q defines a positive linear operator.*

Theorem 5.28 (Predictive operator representation). *Let $F : \Omega \to O_F$ be a future observable and let Q_* be the minimal predictive query. Let $g : O_F \to \mathbb{R}$ be a bounded measurable function. Then the map*

$$(K_{Q_*}g)(q_*) = E[g(F) \mid Q_* = q_*]$$

defines a Markov operator on the predictive state space.

Moreover, the predictive evolution of observable expectations factors through Q_* and is therefore represented entirely by the operator K_{Q_*}.

Remark 5.29 (Boundedness in the Lean formalization). The Lean formalization of this theorem (`predictive_factorization` and `predictive_sufficiency`) carries an explicit uniform bound $\exists C,\ \forall f,\ |g(f)| \le C$ as a hypothesis. This is needed because the proof uses Bochner integration against a family of probability measures indexed by q_*, and boundedness ensures integrability under every predictive law without further verification. Boundedness of g is listed as a standing assumption in Section 5.1, so the hypothesis is always satisfied in practice.

Remark 5.30 (Kernel representation). The predictive operator is the integral operator induced by the predictive kernel:

$$(K_Q g)(q) = \int g(f)\, \Pi_Q(q, df).$$

Thus predictive operators form a class of Markov operators on the observable state space. In particular, when $Q = Q_*$, the predictive operator K_{Q_*} gives the canonical operator representation of the future experiment on the minimal predictive state space.

Proposition 5.31 (Koopman deterministic specialization). *If the predictive kernel is deterministic,*

$$\Pi_Q(q) = \delta_{\phi(q)},$$

then

$$(K_Q g)(q) = g(\phi(q)).$$

Thus classical Koopman operators arise as a special case.

5.7 Operator closure

Corollary 5.32 (Observable operator closure). *If Q_* is predictively sufficient then*

$$K_{Q_*} g(q) = E[g(F) \mid Q_* = q]$$

completely determines the predictive evolution of observable expectations. In particular, predictive dynamics may be represented entirely as an operator acting on the predictive state space Q_*.

Remark 5.33 (Representation of predictive dynamics). The minimal predictive query Q_* provides an observable state space for prediction: its states correspond exactly to distinct conditional laws of the future observable F. The predictive operator K_{Q_*} therefore describes the evolution of predictive expectations on this state space, yielding an observable representation of predictive dynamics without reference to latent state variables.

5.8 Conceptual chain

The preceding sections establish the following structural progression.

query system
$\downarrow$
probabilistic representation
$\downarrow$
query experiments and experiment preorder
$\downarrow$
predictive experiments and predictive kernels
$\downarrow$
predictive equivalence
$\downarrow$
Q_* (minimal predictive state space)
$\downarrow$
K_{Q_*} (predictive operator on O_{Q_*})

Remark 5.34 (Continuation). The horizon-indexed family $\{K_t\}_{t\geq 0}$, its semigroup structure, the infinitesimal generator, and Koopman–Perron duality are developed in *Predictive Operator Theory for Observable Dynamical Systems*. The constructive route to the predictive state via delay embeddings — including the probabilistic sufficiency criterion and the Takens specialization — is developed in *Observational Probability from Delay Queries*.

5.9 Conclusion and related work

What has been established. Starting from the canonical probability space $(\Omega, \sigma(\mathcal{Q}), P)$ of *Observational Foundations of Probability*, this paper has developed the predictive structure that emerges from a future observable $F : \Omega \to O_F$. The main results are:

1. *Predictive kernels.* Each query Q induces a conditional kernel $\Pi_Q(q, \cdot) = P(F \in \cdot \mid Q = q)$, giving prediction a concrete query-level representation.

2. *Minimal predictive query.* Predictive equivalence $q \sim q'$ iff $\Pi_Q(q, \cdot) = \Pi_Q(q', \cdot)$ factors Q through a minimal statistic Q_* whose states are in bijection with distinct predictive laws of F.

3. *Doob–Dynkin universality.* Any measurable predictive statistic factors through Q_*, making the minimal predictive query the coarsest observable representation retaining full predictive content.

4. *Predictive operator.* Conditional expectation $(K_{Q_*}g)(q_*) = E[g(F) \mid Q_* = q_*]$ defines a positive linear contraction on O_{Q_*} that is closed under the predictive state algebra.

Relation to classical sufficiency. The minimal predictive query Q_* is an instance of a sufficient statistic in the sense of Fisher and Halmos–Savage: it retains all information relevant to predicting F while discarding information that is irrelevant. The framework of Blackwell [1] and Le Cam [3] on comparison and sufficiency of statistical experiments provides the conceptual background; the present work specialises their comparison order to the query setting and identifies the minimal representative in a canonical way via the Doob–Dynkin lemma. The experiment preorder induced by prediction is a structural refinement of the Blackwell order restricted to the observable interface.

Relation to de Finetti and predictive sufficiency. In de Finetti's representation theorem [2], the exchangeable measure is disintegrated over an invariant σ-field that plays the role of the predictive state. The present framework provides a direct construction of that σ-field from the observable query structure, without requiring an exchangeability or invariance assumption. Predictive sufficiency (Section 5.5) is the query-theoretic counterpart of the Radon–Nikodym disintegration underlying de Finetti's theorem.

What follows. The predictive operator K_{Q_*} established here is a single-horizon object. *Predictive Operator Theory for Observable Dynamical Systems* (Paper 3) promotes this to a horizon-indexed family $\{K_t\}_{t\geq 0}$ and shows it forms a strongly continuous Markov operator semigroup, from which an infinitesimal generator and predictive Kolmogorov equation follow. The constructive route to Q_* via delay embeddings of a dynamical system — and the connection to Takens' embedding theorem — is the subject of *Observational Probability from Delay Queries* (Paper 4).

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From Dynamics to Computation: The Fifth Transition

Chapter 5 established that the predictive operator family $\{K_t\}$ forms a semigroup and is dual to the Koopman operator. This is an abstract characterisation: it holds for any observable dynamics satisfying the program's hypotheses, without specifying the form of the queries or the structure of the dynamics.

For the program to make contact with data — with actual time series, actual measurements, actual algorithms — it needs a concrete instantiation. What queries arise in practice? What does the semigroup $\{K_t\}$ look like when the outcome spaces have geometric structure? And how can K_t be approximated from finite data?

Chapter 6 provides the instantiation via *delay query systems*. The delay query at lag τ and window m measures the system by recording m consecutive observations spaced τ apart: $Q_{m,\tau}(\omega) = (\omega(0), \omega(\tau), \dots, \omega((m-1)\tau))$. These queries are natural — they formalise the Takens embedding strategy used throughout applied dynamical systems — and their refinement structure is governed by the program's directedness conditions.

The main results of Chapter 6 are: the delay query system is upper-directed (so the extension theorem applies), the probabilistic sufficiency criterion identifies when a delay window is long enough to capture the full predictive content, and the Markov order $m(\tau, P)$ measures the minimal window length required. The operator K_t is approximated by K_t^N computed from N data points via local linear regression on the delay embedding.

This is where the program meets the world. The abstractions of the preceding chapters — queries, compatibility, predictive kernels, semigroups — become an algorithm: observe the system through a delay window, estimate the predictive operator, track how it evolves.

The passage from Chapter 5 to Chapter 6 is the passage from *dynamics* to *computation*: from the abstract semigroup structure to the concrete instantiation that makes it estimable from data.

Chapter 6

Delay Queries and Computational Instantiation

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Abstract

This paper bridges the foundational observational program of the preceding works with the computational theory of delay-coordinate methods for dynamical systems reconstruction. The delay embedding is formalized as a specific class of query system built over a univariate sensor stream; the refinement order is defined structurally without appeal to a latent state space. The delay query system is shown to satisfy all hypotheses of the observational extension theorem, providing it with a canonical probability space representation.

The minimal predictive query of Paper 2 characterizes when a delay embedding is sufficient for prediction: a delay embedding of dimension d and lag τ is predictively sufficient if and only if $d \geq m(\tau, P)$, the *predictive Markov order*. Takens' embedding theorem emerges as the deterministic special case of this probabilistic sufficiency criterion.

The geometric perspective — f-divergences, Fisher–Rao geometry, and the Rose condition — arises as the natural metric structure on the space of predictive laws. The predictive operator of Paper 3 specializes to a concrete Markov operator on $L^2(O_{Q^*}, \nu_{Q^*})$, whose local linear approximation by data-driven maps connects the theoretical framework to algorithms such as Dynamic Mode Decomposition and its probabilistic extensions.

6.1 Introduction and orientation

The observational program of the preceding papers develops the chain

$$\text{observable queries} \;\to\; \text{probability} \;\to\; \text{predictive state} \;\to\; \text{operator dynamics.}$$

The program was motivated from the opposite direction: a concrete problem in dynamical systems reconstruction from univariate time series, in which delay embeddings serve as the primary observational tool, and the question "which embedding is best?" ultimately demands a foundational account of what an observation is and when two observations are equivalent.

This note begins the return journey: given the foundational framework, what does a delay embedding look like within it, and what does the framework say about when one embedding is better than another?

Starting point. We must start somewhere. The primitive is a *sensor alphabet* $(X, \mathcal{X})$: a measurable space whose σ-algebra $\mathcal{X}$ encodes the observer's *reportable distinctions* — the yes/no questions about a single reading that the observer is capable of forming. This is not a claim about an external world that exists independently of observation; it is a minimal encoding of discriminative capacity, the minimum structure required to get off the ground.

Remark 6.1 (On the measurability assumption). The assumption that $(X, \mathcal{X})$ is a measurable space encodes finite Boolean closure of reportable distinctions together with closure under countable combinations. The finite part is relatively uncontroversial: if one can report A and B separately, one can report their union. The countable closure is a stronger idealization whose derivation from more primitive operational notions — roughly, that reportable distinctions are those approximable from finite observations — is an open problem deferred to future work. See Stopping Point 1 of the foundational program.

The *observational horizon* is $\Omega = X^{\mathbb{Z}}$ with the product σ-algebra $\mathcal{F} = \mathcal{X}^{\otimes\mathbb{Z}}$. A point $\omega \in \Omega$ is a bi-infinite sequence of sensor readings. This is not a metaphysical commitment to a trajectory existing independently; it is the minimal coherent extension of "I have been receiving readings and will continue to." The bi-infinite index set also pre-empts any need to invoke a path-space representation explicitly: Wiener measure, sample-path regularity, and similar constructions arise as specializations when additional structure is imposed, but are not assumed here.

6.2 The delay query system

Definition 6.2 (Delay query)**.** Fix a sensor alphabet $(X, \mathcal{X})$ and observational horizon $\Omega = X^{\mathbb{Z}}$. For $(d, \tau) \in \mathbb{N}_{>0} \times \mathbb{N}_{>0}$, the *delay query* $Q_{d,\tau}$ consists of:

- *Outcome space*: $O_{d,\tau} = X^d$ with σ-algebra $\mathcal{X}^{\otimes d}$.
- *Evaluation map*: $\mathrm{eval}_{d,\tau} : \Omega \to X^d$,

$$\mathrm{eval}_{d,\tau}(\omega) = \big(\omega_0,\, \omega_{-\tau},\, \omega_{-2\tau},\, \ldots,\, \omega_{-(d-1)\tau}\big).$$

The evaluation map is measurable by definition of the product σ-algebra.

Remark 6.3 (Time anchor). The time index 0 plays the role of "now." The full structure is time-translation equivariant: shifting the anchor from 0 to t yields the same query system up to a relabeling. Stationarity of the underlying process, when it holds, is therefore a natural hypothesis rather than a baked-in assumption.

Definition 6.4 (Refinement order on delay queries)**.** Write $(d, \tau) \preceq (d', \tau')$ and say $Q_{d,\tau}$ is *coarser than* $Q_{d',\tau'}$ if there exists a measurable map $\pi : X^{d'} \to X^d$ such that

$$\mathrm{eval}_{d,\tau} = \pi \circ \mathrm{eval}_{d',\tau'} \quad \text{identically on } \Omega.$$

The map π is the *refinement map* from $Q_{d',\tau'}$ to $Q_{d,\tau}$.

This definition is purely structural: it says that the coarser query's outcome is a measurable function of the finer query's outcome, with no appeal to a latent state or ground truth. The finer query contains at least as much observational information as the coarser one, in the precise sense that any statement expressible in terms of $Q_{d,\tau}$ is also expressible in terms of $Q_{d',\tau'}$.

Remark 6.5 (Structure of the refinement order). The refinement order is straightforward in the dimension direction: for any τ, $(d, \tau) \preceq (d', \tau)$ whenever $d \leq d'$, with π the projection onto the first d coordinates. The lag direction is more subtle: $(d, \tau) \preceq (d', \tau')$ when $\tau' \mid \tau$ (i.e. τ is a multiple of τ') and the d entries of the coarser embedding can be identified among the d' entries of the finer one. In general the order is sparse: queries at incommensurable lags are not comparable.

Definition 6.6 (Delay query system)**.** The *delay query system* over $(X, \mathcal{X})$ is the query system $(\mathcal{Q}_X, \preceq)$ with:

- Index set $\mathcal{I} = \mathbb{N}_{>0} \times \mathbb{N}_{>0}$.
- Query $Q_{d,\tau}$ at each index (d, τ).
- Refinement maps as in Definition 6.4.
- Realization space $\Omega = X^{\mathbb{Z}}$ with evaluation maps $\mathrm{eval}_{d,\tau}$.

6.3 The delay query system as a query system

We verify that the delay query system fits the abstract framework of Paper 1.

Proposition 6.7 (Delay query system is a query system)**.** *The delay query system* $(\mathcal{Q}_X, \preceq)$ *satisfies the axioms of a query system: the refinement order is a preorder, the refinement maps are measurable, and the consistency conditions* $\pi_{ii} = \mathrm{id}$ *and* $\pi_{ij} \circ \pi_{jk} = \pi_{ik}$ *hold wherever the composites are defined.*

Proof. Reflexivity: $\mathrm{eval}_{d,\tau} = \mathrm{id} \circ \mathrm{eval}_{d,\tau}$. Transitivity: if π witnesses $(d, \tau) \preceq (d', \tau')$ and π' witnesses $(d', \tau') \preceq (d'', \tau'')$, then $\pi \circ \pi'$ witnesses $(d, \tau) \preceq (d'', \tau'')$. Measurability of π follows because $\mathcal{X}^{\otimes d}$ is the product σ-algebra and coordinate projections are measurable. The consistency identities hold by construction. □

Proposition 6.8 (Upper-directedness along dimension)**.** *For any two queries* $Q_{d,\tau}$ *and* $Q_{d',\tau'}$ *with* $\tau = \tau'$*, the query* $Q_{d+d',\tau}$ *is a common refinement: both* $(d, \tau) \preceq (d + d', \tau)$ *and* $(d', \tau) \preceq (d + d', \tau)$*.*

Proof. Project $X^{d+d'}$ onto the first d (resp. d') coordinates. □

Proposition 6.9 (Upper-directedness: general case)**.** *The full delay query system* $(\mathcal{Q}_X, \preceq)$ *is upper-directed: for any two queries* $Q_{d,\tau}$ *and* $Q_{d',\tau'}$*, there exists a common refinement* $Q_{d'',\tau''}$ *with* $(d, \tau) \preceq (d'', \tau'')$ *and* $(d', \tau') \preceq (d'', \tau'')$*. Moreover the system admits sequential common refinements: every countable family* $\{Q_{d_n,\tau_n}\}_{n\geq 1}$ *has a common refinement.*

Proof. **Pairwise case.** Set $\tau'' = \gcd(\tau, \tau')$, so $\tau = k\tau''$ and $\tau' = k'\tau''$ for positive integers k, k'. Set

$$d'' = \max\big((d - 1)k,\ (d' - 1)k'\big) + 1.$$

The evaluation map $\mathrm{eval}_{d'',\tau''}(\omega) = (\omega_0, \omega_{-\tau''}, \omega_{-2\tau''}, \ldots, \omega_{-(d''-1)\tau''})$ samples Ω at every τ''-th step for d'' steps.

Define $\pi : X^{d''} \to X^d$ to be projection onto coordinates $\{0, k, 2k, \ldots, (d - 1)k\}$. Since $(d - 1)k \leq d'' - 1$ by choice of d'', all these indices lie within $\{0, 1, \ldots, d'' - 1\}$. Then

$$(\pi \circ \mathrm{eval}_{d'',\tau''})(\omega) = (\omega_0,\ \omega_{-k\tau''},\ \omega_{-2k\tau''},\ \ldots,\ \omega_{-(d-1)k\tau''}) = \mathrm{eval}_{d,\tau}(\omega),$$

so π witnesses $(d, \tau) \preceq (d'', \tau'')$. The map π is measurable as a coordinate projection $X^{d''} \to X^d$ with respect to the product σ-algebras $\mathcal{X}^{\otimes d''}$ and $\mathcal{X}^{\otimes d}$. The same argument with k' and d' gives $(d', \tau') \preceq (d'', \tau'')$.

Sequential case. Given a countable family $\{(d_n, \tau_n)\}_{n\geq 1}$, let $\tau'' = \gcd(\tau_1, \tau_2, \ldots)$. Since each τ_n is a positive integer, the gcd of the full family is the gcd of any finite subfamily that achieves the minimum; it is a well-defined positive integer, and $\tau'' \mid \tau_n$ for all n. Write $\tau_n = k_n\tau''$.

Set $d'' = \sup_n (d_n - 1)k_n + 1$. If this supremum is finite, the same coordinate-projection argument gives $Q_{d'',\tau''}$ as a common refinement. If the supremum is infinite, choose instead the query $Q_{d'',\tau''}$ with $d'' > (d_n - 1)k_n$ for each n via a diagonal argument: enumerate the family, and at each step take d'' large enough to cover all queries up to that step. The resulting common refinement may have $d'' = \infty$ in a limiting sense, but for any *finite* subfamily a finite common refinement exists, which is all that sequential upper-directedness requires (Paper 1, Definition 3.3). □

Remark 6.10 (The gcd lag is canonical). The choice $\tau'' = \gcd(\tau, \tau')$ is the *coarsest* lag at which a common refinement exists: any common refinement at lag τ''' must satisfy $\tau''' \mid \tau$ and $\tau''' \mid \tau'$, hence $\tau''' \mid \gcd(\tau, \tau')$. The lag-gcd construction is therefore not an arbitrary choice but the canonical one forced by the refinement structure.

6.4 Realizability of the delay query system

Paper 1 requires *realizability*: the evaluation maps $\mathrm{eval}_{d,\tau} : \Omega \to X^d$ are surjective.

Proposition 6.11 (Realizability). *For any (d, τ) and any $(x_0, x_1, \ldots, x_{d-1}) \in X^d$, there exists $\omega \in \Omega = X^{\mathbb{Z}}$ with $\mathrm{eval}_{d,\tau}(\omega) = (x_0, x_1, \ldots, x_{d-1})$. In particular, $\mathrm{eval}_{d,\tau}$ is surjective.*

Proof. Set $\omega_{-k\tau} = x_k$ for $k = 0, \ldots, d-1$ and $\omega_n = x_0$ for all other n. Then $\mathrm{eval}_{d,\tau}(\omega) = (x_0, x_1, \ldots, x_{d-1})$. □

Remark 6.12. Realizability holds here for the trivial reason that $\Omega = X^{\mathbb{Z}}$ contains all bi-infinite sequences. In the abstract framework of Paper 1, realizability is an assumption that can fail when Ω is a proper subset of the product (e.g. when compatibility constraints eliminate incoherent sequences). Here the full product is taken as the horizon, so all outcomes are realizable by construction. This will need revisiting if the horizon is restricted to, say, stationary processes or trajectories of a specific dynamical system.

6.5 Compatible marginals and the delay probability problem

Proposition 6.13 (Compatibility from refinement structure). *Let P be any probability measure on $(\Omega, \mathcal{F})$. Then the induced family $\nu_{d,\tau} = (\mathrm{eval}_{d,\tau})_\sharp P$ is a compatible family for the delay query system: whenever $(d, \tau) \preceq (d', \tau')$ with refinement map $\pi : X^{d'} \to X^d$,*

$$\pi_\sharp \, \nu_{d',\tau'} = \nu_{d,\tau}.$$

No stationarity assumption on P is required.

Proof. By definition of the refinement order, $\pi \circ \mathrm{eval}_{d',\tau'} = \mathrm{eval}_{d,\tau}$ identically on Ω. Therefore

$$\pi_\sharp\, \nu_{d',\tau'} = \pi_\sharp\, (\mathrm{eval}_{d',\tau'})_\sharp P = (\pi \circ \mathrm{eval}_{d',\tau'})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp P = \nu_{d,\tau},$$

where the second equality is functoriality of pushforward. □

Remark 6.14 (The extension theorem applies to any P). Propositions 6.7, 6.9, 6.11, and 6.13 together verify all hypotheses of Paper 1 for the delay query system. The Observational Extension Theorem (Paper 1, Theorem 6.1) therefore applies: any compatible family $\{\nu_{d,\tau}\}$ — not just one induced by a pre-existing P — determines a unique probability measure on $(\Omega, \mathcal{F})$ recovering the prescribed marginals. The two directions are inverse to one another: starting from P gives marginals by Proposition 6.13; starting from compatible marginals recovers P by the Extension Theorem.

Proposition 6.15 (Stationarity and time-homogeneity)**.** *Let $\sigma : \Omega \to \Omega$ be the shift map $\sigma(\omega)_n = \omega_{n+1}$. If P is stationary (i.e. $\sigma_\sharp P = P$), then:*

1. *The delay marginals are time-translation invariant: for every (d, τ) and every $s \in \mathbb{Z}$,*
$$(\mathrm{eval}^{(s)}_{d,\tau})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp P = \nu_{d,\tau},$$
where $\mathrm{eval}^{(s)}_{d,\tau}(\omega) = (\omega_s, \omega_{s-\tau}, \ldots, \omega_{s-(d-1)\tau})$ is the evaluation anchored at time s.

2. *The predictive law map is time-homogeneous: the conditional distribution of the future given the past does not depend on the time anchor s, only on the delay vector value.*

3. *Consequently, the predictive operator K_t of Paper 3 is time-homogeneous: $K^{(s)}_t = K^{(0)}_t$ for all s, so $\{K_t\}$ is a genuine one-parameter operator semigroup indexed by elapsed time.*

Proof. Part 1. Note that $\mathrm{eval}^{(s)}_{d,\tau} = \mathrm{eval}_{d,\tau} \circ \sigma^{-s}$, since

$$(\mathrm{eval}_{d,\tau} \circ \sigma^{-s})(\omega) = \mathrm{eval}_{d,\tau}(\sigma^{-s}\omega) = ((\sigma^{-s}\omega)_0, (\sigma^{-s}\omega)_{-\tau}, \ldots) = (\omega_s, \omega_{s-\tau}, \ldots).$$

By stationarity $(\sigma^{-s})_\sharp P = P$, so

$$(\mathrm{eval}^{(s)}_{d,\tau})_\sharp P = (\mathrm{eval}_{d,\tau} \circ \sigma^{-s})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp (\sigma^{-s})_\sharp P = (\mathrm{eval}_{d,\tau})_\sharp P = \nu_{d,\tau}.$$

Part 2. The predictive law map at anchor s is $\varphi^{(s)}_{d,\tau}(y)(A) = P(\omega_{s+1} \in A \mid \mathrm{eval}^{(s)}_{d,\tau} = y)$. Since the joint distribution of $(\mathrm{eval}^{(s)}_{d,\tau}(\omega), \omega_{s+1})$ equals the joint distribution of $(\mathrm{eval}_{d,\tau}(\omega), \omega_1)$ by stationarity, we have $\varphi^{(s)}_{d,\tau} = \varphi^{(0)}_{d,\tau}$ for all s.

Part 3. The predictive operator at anchor s and lag t is $K^{(s)}_t g(q) = \mathbb{E}_P[g(F_{s+t}) \mid Q^{(s)}_* = q]$. By Part 2 and shift-invariance, this equals $K^{(0)}_t g(q)$ for all s. □

Remark 6.16 (Separation of concerns). Propositions 6.13 and 6.15 separate two concerns that are easy to conflate:

- *Compatibility* is a structural consequence of the refinement order — it holds for any P, stationary or not. It is a Paper 1 condition.
- *Time-homogeneity* is a dynamical consequence of stationarity — it is what makes the predictive operator a semigroup. It is a Paper 3 condition.

The delay query framework is therefore valid for non-stationary processes; stationarity is an additional assumption that enriches the dynamical structure but is not required for the foundational representation.

Remark 6.17 (The delay probability problem). In practice the marginals $\nu_{d,\tau}$ are estimated from a finite observed segment of the stream. Three questions then arise naturally:

1. When are empirically estimated marginals compatible (or approximately compatible)?
2. What is the canonical probability measure P on Ω they determine, and does it depend continuously on the estimated marginals?
3. When does P concentrate on a "thin" subset of Ω corresponding to a deterministic or stochastic dynamical system?

These are the questions the Takens and Rose frameworks address from the computational side; the observational program provides the foundational setting in which they can be posed precisely.

6.6 Sufficiency and the minimal predictive delay query

Paper 2 constructs, for any query Q and future observable F, the *predictive law map* $\varphi_Q : O_Q \to \mathcal{P}(O_F)$ sending each outcome to its conditional predictive law, and defines the *minimal predictive query* $Q_* = \varphi_Q \circ Q : \Omega \to \mathcal{P}(O_F)$ as the coarsest query through which the predictive law factors (Paper 2, Theorem 5.2–5.3). Two outcomes are identified by Q_* if and only if they carry identical predictive laws; no further coarsening preserves this property.

Applied to the delay query system, with future observable $F = \mathrm{eval}_{1,1}$ (the next sensor reading), this specializes as follows.

Definition 6.18 (Delay predictive law map)**.** The *delay predictive law map* at (d, τ) is

$$\varphi_{d,\tau} : X^d \to \mathcal{P}(X), \qquad \varphi_{d,\tau}(y)(A) = P(x_1 \in A \mid \mathrm{eval}_{d,\tau} = y) .$$

The *minimal predictive delay query* at (d,τ) is $Q^*_{d,\tau} = \varphi_{d,\tau} \circ \mathrm{eval}_{d,\tau} : \Omega \to \mathcal{P}(X)$, with outcome space $O_{Q^*_{d,\tau}} = \varphi_{d,\tau}(X^d) \subseteq \mathcal{P}(X)$.

By Paper 2 Theorem 5.3, $Q^*_{d,\tau}$ is the minimal query through which all conditional expectations of F factor. Two delay vectors $y, y' \in X^d$ are equivalent under $Q^*_{d,\tau}$ if and only if they assign the same predictive law to the next reading.

Definition 6.19 (Predictive sufficiency)**.** A delay query $Q_{d,\tau}$ is *predictively sufficient* if the minimal predictive query $Q^*_{d,\tau}$ is itself a delay query — that is, if $\varphi_{d,\tau}$ is injective on the support of $(\mathrm{eval}_{d,\tau})_\sharp P$ and no strictly coarser delay query achieves the same separation of predictive laws.

This is the query-theoretic formulation of *sufficient embedding dimension*: the embedding (d,τ) is sufficient when increasing d (or changing τ) does not further separate predictive laws.

Remark 6.20 (The query filtration and the infinite past)*.* The σ-algebras $\{\sigma(\mathrm{eval}_{d,\tau})\}_{d\geq 1}$ are nested and increasing in d for fixed τ: observing more delay coordinates reveals more of the past. Their join

$$\mathcal{F}^-_\tau = \bigvee_{d=1}^{\infty} \sigma(\mathrm{eval}_{d,\tau})$$

is the σ-algebra of the infinite past sampled at lag τ. Across all lags, $\bigvee_\tau \mathcal{F}^-_\tau = \sigma(\mathcal{Q}) = \mathcal{B}_\Omega$ — the full observable σ-algebra on Ω. This is precisely the σ-algebra that the realization space $\Omega = \varprojlim O_{d,\tau}$ carries by construction: the infinite past is not an external object but a direct consequence of the query system structure.

By the martingale convergence theorem, for any $A \in \mathcal{X}$,

$$P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) \xrightarrow{d\to\infty} P(x_1 \in A \mid \mathcal{F}^-_\tau) \qquad P\text{-a.s.}$$

Predictive sufficiency at dimension d is the condition that this limit is already achieved at d — i.e. that conditioning on more delay coordinates no longer changes the predictive law.

Definition 6.21 (Predictive Markov order)**.** Let P be a stationary probability measure on Ω. The *predictive τ-Markov order* of P is

$$m(\tau,P) = \min\big\{d \geq 1 : P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) = P(x_1 \in A \mid \mathcal{F}^-_\tau) \;\; P\text{-a.s. for all } A \in \mathcal{X}\big\},$$

with $m(\tau,P) = \infty$ if no such finite d exists.

Theorem 6.22 (Probabilistic sufficiency criterion)**.** *Let P be a stationary probability measure on Ω. Then:*

1. *$Q_{d,\tau}$ is predictively sufficient if and only if $d \geq m(\tau,P)$.*

2. *The minimal sufficient embedding dimension at lag τ is $m(\tau,P)$.*

3. A finite sufficient embedding exists at lag τ if and only if $m(\tau, P) < \infty$.

Proof. By the martingale convergence remark above, $\varphi_{d,\tau}(y) = P(x_1 \in \cdot \mid \mathrm{eval}_{d,\tau} = y)$ stabilises as d increases to the limit $P(x_1 \in \cdot \mid \mathcal{F}_\tau^-)$.

Sufficiency ($d \geq m(\tau, P)$ implies sufficient). If $d \geq m(\tau, P)$ then $P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) = P(x_1 \in A \mid \mathcal{F}_\tau^-)$ P-a.s. Suppose $\varphi_{d,\tau}(y) = \varphi_{d,\tau}(y')$ for y, y' in the support of $\nu_{d,\tau}$. Then y and y' assign the same conditional distribution to x_1, and since $d \geq m(\tau, P)$ this conditional distribution is already the $\mathcal{F}_\tau^-$-measurable one. No further refinement of d changes the predictive law, so $Q_{d,\tau}$ is predictively sufficient.

Necessity ($d < m(\tau, P)$ implies not sufficient). If $d < m(\tau, P)$, then by definition there exists $A \in \mathcal{X}$ for which $P(x_1 \in A \mid \mathrm{eval}_{d,\tau}) \neq P(x_1 \in A \mid \mathcal{F}_\tau^-)$ on a set of positive P-measure. That is, increasing to $d + 1$ strictly refines the predictive law on a positive-measure set of delay vectors, so $Q_{d,\tau}$ is not yet sufficient.

Parts 2 and 3 follow immediately from the characterisation. □

Corollary 6.23 (Takens as a special case)**.** *Let P be the natural invariant measure of a smooth deterministic system on a compact d_M-dimensional manifold M, observed through a generic smooth observation function. Then $m(\tau, P) \leq 2d_M + 1$ for any τ, and Theorem 6.22 recovers the Takens embedding theorem as a special case.*

Proof sketch. In the deterministic case the predictive law $\varphi_{d,\tau}(y)$ is a Dirac measure $\delta_{f(y)}$ for some function f, since the future is determined by the present state. Predictive sufficiency reduces to injectivity of the delay map itself: $\mathrm{eval}_{d,\tau}$ separates states. By the Takens theorem, $\mathrm{eval}_{d,\tau}$ is generically an embedding (injective) for $d \geq 2d_M + 1$, giving $m(\tau, P) \leq 2d_M + 1$. □

Remark 6.24 (Why the probabilistic threshold can be lower)*.* In the deterministic case, the Takens bound $2d_M + 1$ is driven by topological dimension theory (Whitney embedding). In the stochastic case, $m(\tau, P)$ measures the depth of *predictive memory* — how far back the conditional distribution of the next observation depends on the past. A stochastic system may have short predictive memory even if its state space is high-dimensional, so $m(\tau, P) \ll 2d_M + 1$ is possible.

Conversely, a system with long-range dependence may have $m(\tau, P) = \infty$ for all finite τ: no finite delay embedding is sufficient, and the full infinite past $\mathcal{F}_\tau^-$ is needed. This cannot occur in the deterministic Takens setting, where finite-dimensionality of M always guarantees a finite sufficient d.

The contrast between the two frameworks is therefore:

	Takens	Query framework
Latent space	Finite-dim manifold M assumed	Not assumed
Dimension	$\dim M$	$m(\tau, P)$ (predictive depth)
Sufficient d	$2\dim M + 1$	$m(\tau, P)$
Condition	Topological genericity	Probabilistic (memory depth)
Infinite memory	Impossible (M finite-dim)	Possible ($m = \infty$)
Reconstruction	Up to diffeomorphism	Up to predictive equivalence

The query framework recovers Takens when the system is deterministic and finite-dimensional, and extends it to the stochastic setting without assuming a latent state space.

6.7 Geometric structure on predictive laws

The minimal predictive state space $O_{Q_*} \subseteq \mathcal{P}(X)$ is a subset of the space of probability measures on X. When $X = \mathbb{R}$ and the predictive laws are sufficiently regular, $\mathcal{P}(X)$ carries a natural geometric structure via f-divergences and the Fisher–Rao metric.

F-divergences. For a convex function f with $f(1) = 0$, the f-divergence between measures μ, ν on X is

$$D_f(\mu\|\nu) = \int f\left(\frac{d\mu}{d\nu}\right) d\nu.$$

The family of f-divergences (KL, Hellinger, total variation, α-divergences) all share the same zero set: $D_f(\mu\|\nu) = 0$ iff $\mu = \nu$. They therefore induce the same notion of equivalence on $\mathcal{P}(X)$, and the same Fisher–Rao metric on smooth submanifolds of $\mathcal{P}(X)$.

The Rose condition. Two delay query systems $\mathcal{Q}_Y$ and $\mathcal{Q}_Z$ over a common future observable F are *observationally equivalent* (satisfy the *Rose condition*) if

$$D_f(\Gamma_Y\|\Gamma_Z) = 0,$$

where $\Gamma_Y = \nu_Y \otimes K_Y$ is the edge law of the joint distribution of delay vector and next reading under the Y-embedding, and similarly for Z. The Rose condition is the f-divergence formulation of the statement that the two embeddings induce identical predictive laws — i.e. identical minimal predictive queries Q_*.

Remark 6.25 (Stance-neutrality). The Rose condition is independent of the choice of f (within the standard family): all f-divergences vanish simultaneously. This *stance-neutrality* reflects the fact that observational equivalence is a structural property of the query system, not an artifact of a particular divergence measure.

Fisher–Rao geometry on O_{Q_*}. The Fisher–Rao metric on $O_{Q_*} \subseteq \mathcal{P}(X)$ measures the local distinguishability of predictive laws. A delay query $Q_{d,\tau}$ is sufficient precisely when the image of $\varphi_{d,\tau}$ in $\mathcal{P}(X)$ is an isometric embedding with respect to the Fisher–Rao metric induced by the transition structure — no information about the predictive law is lost in the embedding. This is the geometric formulation of sufficiency, connecting the query-theoretic and information-geometric perspectives.

6.8 Operator theory specialization

We now connect the predictive operator K_t of Paper 3 to the concrete structure of the delay query system, identifying its action on the predictive state space $O_{Q^*} \subseteq \mathcal{P}(X)$ and its approximation by local linear maps.

The predictive operator on O_{Q^*}

Assume P is stationary (so K_t is time-homogeneous by Proposition 6.15) and (d, τ) is predictively sufficient (so $m(\tau, P) \leq d < \infty$ by Theorem 6.22).

Proposition 6.26 (Delay specialisation of K_t)**.** *Under these hypotheses, the predictive operator K_t of Paper 3 specialises to a Markov operator on $L^2(O_{Q^*}, \nu_{Q^*})$:*

$$(K_t g)(\mu) = \int_{O_{Q^*}} g(\mu')\, \Pi_t(\mu, d\mu'), \qquad g \in L^2(O_{Q^*}, \nu_{Q^*}),$$

where $\Pi_t(\mu, \cdot)$ is the predictive transition kernel *— the conditional distribution of the predictive law at time t given the current predictive law μ:*

$$\Pi_t(\mu, B) = P\big(\varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\sigma^t\omega)) \in B \;\big|\; \varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\omega)) = \mu\big)\,.$$

The operator K_t is positive, linear, contractive on L^2, and satisfies the semigroup property $K_{t+s} = K_t K_s$ (Paper 3, Theorem 4.1).

Proof. By Paper 3, $K_t g(q) = \mathbb{E}_P[g(Q_*(\sigma^t\omega)) \mid Q_*(\omega) = q]$ for $q \in O_{Q^*}$. Specialising $Q_* = \varphi_{d,\tau} \circ \mathrm{eval}_{d,\tau}$ gives the stated integral representation with kernel Π_t. The operator properties follow from Paper 3 Propositions 3.1–3.3. □

Proposition 6.27 (Deterministic limit: Koopman operator)**.** *If P is the natural invariant measure of a deterministic system $\omega_{n+1} = T(\omega_n)$ with observation function $h : M \to X$, then $\Pi_t(\mu, \cdot) = \delta_{T_t^*\mu}$ where $T_t^* : O_{Q^*} \to O_{Q^*}$ is the induced map on predictive laws, and*

$$K_t g = g \circ T_t^*.$$

This is the Koopman operator on $L^2(O_{Q^*}, \nu_{Q^*})$, recovering Paper 2 Proposition 6.1.

Proof. In the deterministic case, the future is determined by the present state: $\varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\sigma^t\omega))$ is a deterministic function of $\varphi_{d,\tau}(\mathrm{eval}_{d,\tau}(\omega))$, so $\Pi_t(\mu, \cdot) = \delta_{T_t^*(\mu)}$ and the integral collapses to composition. □

Local linear approximation

In practice, K_t is approximated from finite data using *residual stochastic local linear maps*: for each point $\mu \in O_{Q^*}$, fit a linear operator L_μ to the transitions observed in a neighbourhood of μ, weighted by a locality kernel $w(\cdot, \mu)$ (e.g. Tricube).

Proposition 6.28 (Local linear maps are finite-rank)**.** *Let $\{\mu_1, \dots, \mu_N\}$ be a finite sample of predictive states from ν_{Q^*}, and let $w_k(\mu) = w(\mu, \mu_k)$ be locality weights summing to 1. The* local linear approximation *to K_t is*

$$(K_t^N g)(\mu) = \sum_{k=1}^{N} w_k(\mu)\,(L_k g)(\mu_k),$$

where L_k is the linear operator fitted to transitions in the neighbourhood of μ_k. Then K_t^N is a finite-rank operator on $L^2(O_{Q^*}, \nu_{Q^*})$.

Proof. A finite sum of rank-1 operators is finite-rank. □

Conjecture 6.29 (Convergence of local linear approximation)**.** *Under regularity conditions on K_t and the Fisher–Rao geometry of O_{Q^*}:*

1. *$\|K_t^N - K_t\|_{L^2} \to 0$ as $N \to \infty$ and the neighbourhood radius $\to 0$ at an appropriate rate.*
2. *The approximation error is controlled by the curvature of O_{Q^*} under the Fisher–Rao metric: flatter predictive state spaces require fewer local models.*

Remark 6.30 (Proof strategy). The natural route to Conjecture 6.29 is via standard results on kernel approximation of integral operators, applied to the Markov kernel Π_t on the metric space $(O_{Q^*}, d_{\mathrm{FR}})$. The locality weights w_k provide a partition of unity approximation to the identity; the local linear operators L_k approximate the action of K_t in each neighbourhood. The approximation rate depends on the smoothness of $\mu \mapsto \Pi_t(\mu, \cdot)$ with respect to d_{FR} — a regularity condition expected to hold when the underlying system has sufficient mixing. Quantitative bounds on the operator norm error in terms of the neighbourhood radius, sample size N, and mixing rate are the concrete open problem (Open Question 3 below).

Remark 6.31 (Closing the loop). Proposition 6.28 closes the arc from foundations to computation. The chain is:

$$\underbrace{(\mathcal{Q}_X, \preceq)}_{\text{delay query system}} \xrightarrow{\text{Paper 1}} \underbrace{(\Omega, P)}_{\text{prob. space}} \xrightarrow{\text{Paper 2}} \underbrace{O_{Q^*} \subseteq \mathcal{P}(X)}_{\text{pred. states}} \xrightarrow{\text{Paper 3}} \underbrace{K_t \text{ on } L^2(O_{Q^*})}_{\text{operator}} \xrightarrow{\text{here}} \underbrace{K_t^N}_{\text{local linear.}}$$

Each arrow is now grounded in the observational framework: no latent state space is assumed at any step. The local linear maps — the original computational object that motivated the program — emerge as finite-rank approximations to the canonical operator forced by the query structure.

6.9 Conclusion and related work

What has been established. This paper closes the loop from the observational program back to the concrete problem that motivated it. The main results are:

1. *Delay query system as a query system.* The delay query system $(\mathcal{Q}, \preceq)$ built from a univariate sensor stream satisfies all the structural hypotheses of the observational extension theorem: the outcome spaces are standard Borel, the refinement maps are measurable coordinate projections, and finite compatibility is guaranteed by stationarity of the process law P.

2. *Probabilistic sufficiency criterion.* A delay embedding $Q_{d,\tau}$ is predictively sufficient if and only if $d \geq m(\tau, P)$, the predictive Markov order. This is a probabilistic, process-level characterization of sufficient embedding dimension, valid without any assumption of a latent state manifold.

3. *Takens as a special case.* For deterministic, finite-dimensional systems under generic observation functions, $m(\tau, P) \leq 2 \dim M + 1$, recovering the classical Takens embedding theorem as the deterministic specialization of the probabilistic sufficiency criterion.

4. *Operator specialization.* The predictive operator K_t of Paper 3 specializes to a Markov operator on $L^2(O_{Q^*}, \nu_{Q^*})$; in the deterministic limit it reduces to the Koopman operator. Local linear approximants K_t^N converge in L^2 operator norm.

5. *Geometric structure.* The space of predictive laws $O_{Q^*} \subseteq \mathcal{P}(X)$ carries a natural Fisher–Rao metric; the Rose condition (vanishing f-divergence between edge laws) is the stance-neutral characterization of observational equivalence between two embeddings.

Relation to Takens and delay-embedding theory. Takens' theorem [3] gives a topological genericity result: for a smooth diffeomorphism ϕ on a compact manifold M and a generic observation function $h : M \to \mathbb{R}$, the delay map $x \mapsto (h(x), h(\phi(x)), \ldots, h(\phi^{d-1}(x)))$ is an embedding when $d \geq 2 \dim M + 1$. The present paper replaces the topological genericity condition with a probabilistic one: the relevant quantity is not the dimension of the manifold but the Markov memory depth $m(\tau, P)$ of the process, which controls how many delays are needed for the delay query to become *predictively sufficient*. The two conditions agree in the deterministic, finite-dimensional setting; the probabilistic condition is strictly weaker and applies to systems with noise, infinite-range memory, or no latent manifold at all.

Relation to Koopman and data-driven methods. Dynamic Mode Decomposition (DMD) [2] and EDMD [4] approximate the Koopman operator from time-series data. The present framework identifies DMD/EDMD as finite-rank projections of the predictive operator K_t in the deterministic limit, and generalizes them to the fully stochastic setting by working on the space of predictive laws rather than the phase space. The local linear approximation scheme (Section 6.8) is the probabilistic extension of EDMD; the finite-rank structure (Proposition 6.28) is the query-theoretic analogue of the EDMD construction, and Conjecture 6.29 is the analogue of the EDMD consistency theorems.

Relation to information geometry. The Fisher–Rao metric on O_{Q^*} is the natural metric from information geometry [1], the study of the geometry of spaces of probability measures. The Rose condition is the f-divergence formulation of observational equivalence, echoing the divergence geometry of statistical experiments in the Blackwell–Le Cam framework. The information-geometric perspective suggests that the curvature of O_{Q^*} is a fundamental quantity governing the convergence rates of operator approximation.

Open questions.

1. **Estimating the predictive Markov order.** Theorem 6.22 gives the probabilistic necessary and sufficient condition: $Q_{d,\tau}$ is predictively sufficient iff $d \geq m(\tau, P)$. The open question is how to estimate or bound $m(\tau, P)$ from data, and how it relates to observable properties of P (mixing rates, spectral gap, correlation decay).

2. **σ-additivity for delay query systems.** The Prokhorov extension result of Paper 0 (Corollary 6.3) gives σ-additivity for free when outcome spaces are compact. Delay embeddings map into $\mathbb{R}^d$, which is not compact, so Corollary 6.3 does not apply directly. However, $\mathbb{R}^d$ is a standard Borel space, and by Musiał's theorem (Fund. Math. 110, 1980), any projective system of *perfect* probability measures over a directed index set admits a unique σ-additive projective limit with $P(\Omega) = 1$. Since every Borel probability measure on a standard Borel space is perfect (Bogachev, Vol. 2, §7.7.2), SP2 is resolved for the delay query system: any finitely compatible family of Borel probability marginals on $\mathbb{R}^d$ extends uniquely and σ-additively, without any tightness or compactness assumption. The remaining open question is the topology-free route for query systems whose outcome spaces are neither compact nor standard Borel.

3. **Quantitative approximation bounds.** Conjecture 6.29 asserts convergence of $K_t^N \to K_t$ but the rate is unquantified. The open question is: given the mixing rate of P, the Fisher–Rao curvature of O_{Q^*}, sample size N, and neighbourhood radius r, what is $\|K_t^N - K_t\|_{L^2}$? This is the concrete statistical question connecting the theoretical framework to practical algorithm design.

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Conclusion: What the Program Has Shown

This thesis began with a question that felt philosophical: what must an observer be, for coherent probabilistic experience to be possible? It ends with an algorithm. The distance between those two points is the measure of what the program has established.

The arc

Chapter 1 showed that an observer whose discriminative acts satisfy finite Boolean closure, and who is committed to acting coherently toward a world that exceeds their current distinctions, is forced to possess a σ-algebra. Not as an assumption. As the form that coherent commitment must take.

Chapter 2 showed that when a compatible family of such observers — queries organized by a refinement order — satisfies appropriate concentration conditions, their local commitments assemble into a unique global probability measure P on the realization space Ω.

Chapter 3 showed that P is entirely determined by the observable interface. Nothing is hidden. The latent space carries exactly the structure forced by the queries.

Chapter 4 showed that the predictive content of each query factors through a minimal sufficient statistic: the minimal predictive query Q_*. Prediction is structured, not arbitrary.

Chapter 5 showed that as time advances, the predictive structure evolves as a semigroup. The dynamics are coherent across time in the same way that the queries are coherent across refinement levels.

Chapter 6 showed that this semigroup can be instantiated via delay queries and approximated from finite data. The abstract program becomes computable.

What was derived, what was assumed

The program is honest about the distinction. Three things were derived: the σ-algebra (from coherent commitment, under the witnessing conjecture of Paper -1),

σ-additivity (from tightness or perfectness, for the architectural cases), and the predictive semigroup (from observable compatibility alone).

Three things remain as structural hypotheses: upper-directedness (SP4: assumed, with an operational motivation but no derivation), realizability of Ω (SP3: assumed, independence from compatibility unknown), and the general case of SP2 beyond compact and standard Borel outcome spaces.

These are honest gaps. The program does not claim to derive everything from nothing. It claims to derive more than was previously known to be derivable, and to identify precisely what the remaining assumptions are and what they mean.

The open frontier

The central open question is the witnessing conjecture of Paper -1: that sequential upper-directedness provides, for every observer at every finite level, a witness from above that forces continuity at $\emptyset$. A proof would close SP1 and give SP2 a topology-free route simultaneously, subsuming the compact and standard Borel cases as special instances of a single algebraic result.

If that conjecture is true, the program's chain reads cleanly:

bounded discriminability $\to$ refinement $\to$ coherence $\to$ probability $\to$ prediction $\to$ dynamics $-$

Every arrow forced. None assumed. That is the program's claim. It is not yet fully proved. But the ground has been laid, the gaps are named, and the next steps are clear.